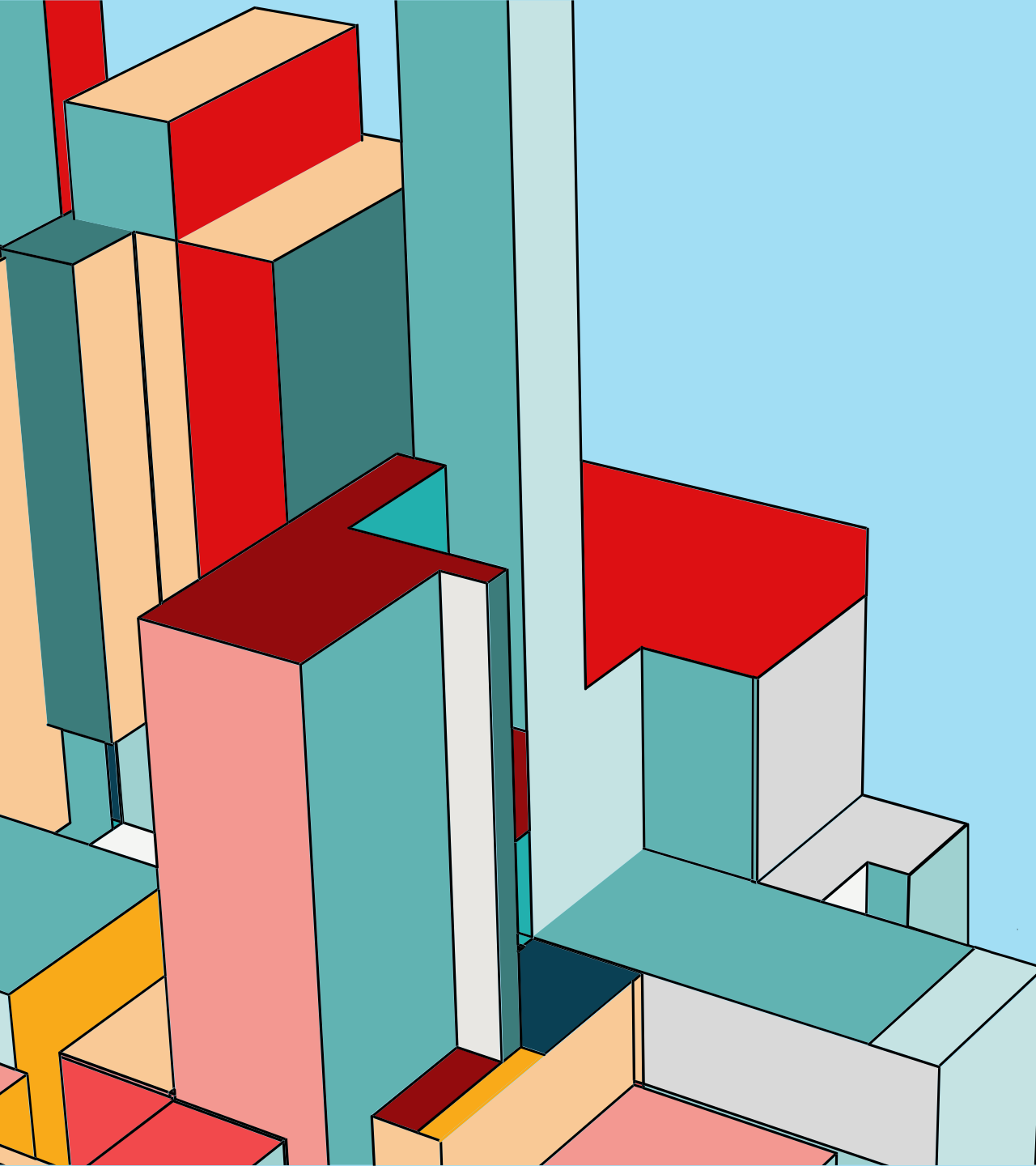




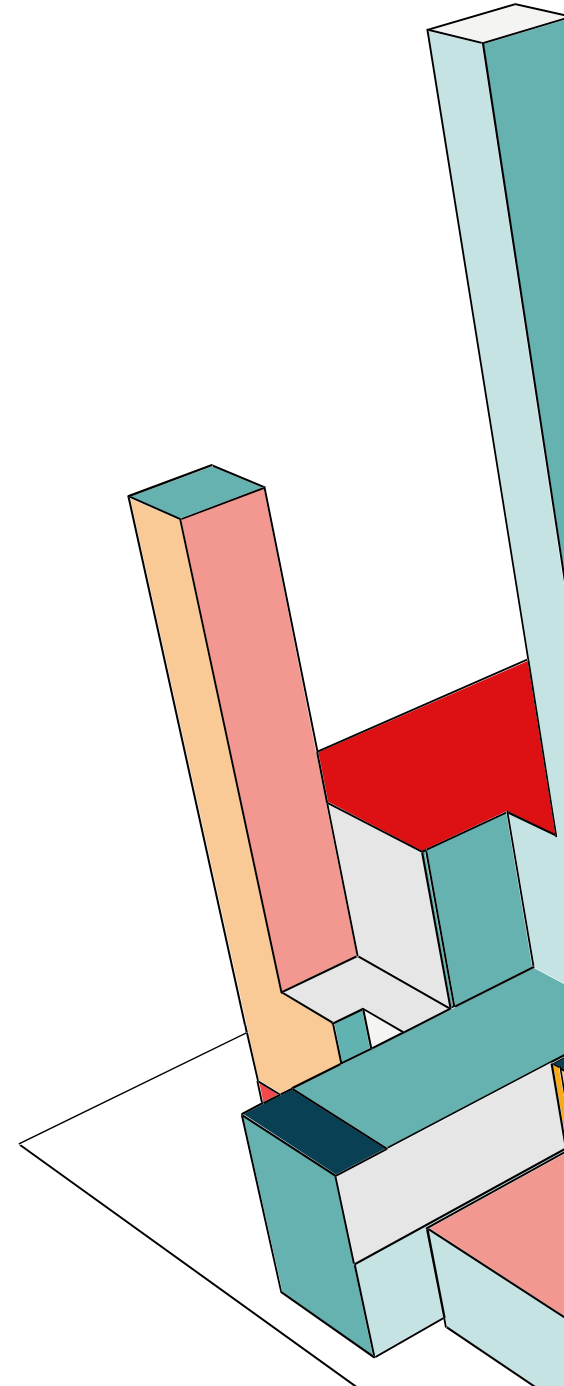
CREDIT EDA ASSIGNMENT

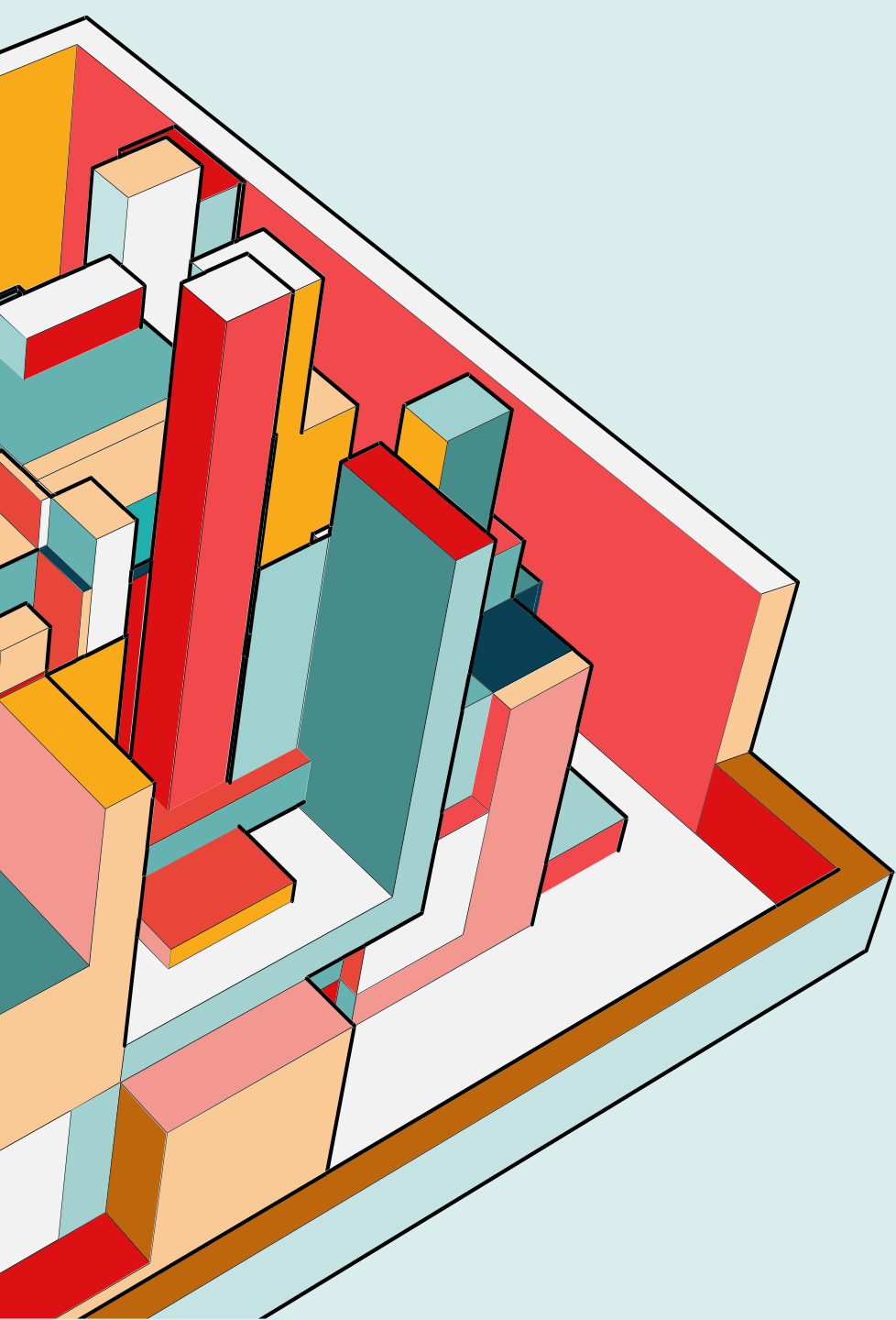
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HOME CREDIT CASE

AGENDA

1. Introduction
2. Data Quality and Feature Processing
3. Analysis of Previous Applications
4. Univariate Analysis
5. Segmented Univariate Analysis
6. Correlation (Numeric) Analysis
7. Bivariate Analysis
8. Insights Summary
9. Recommendations

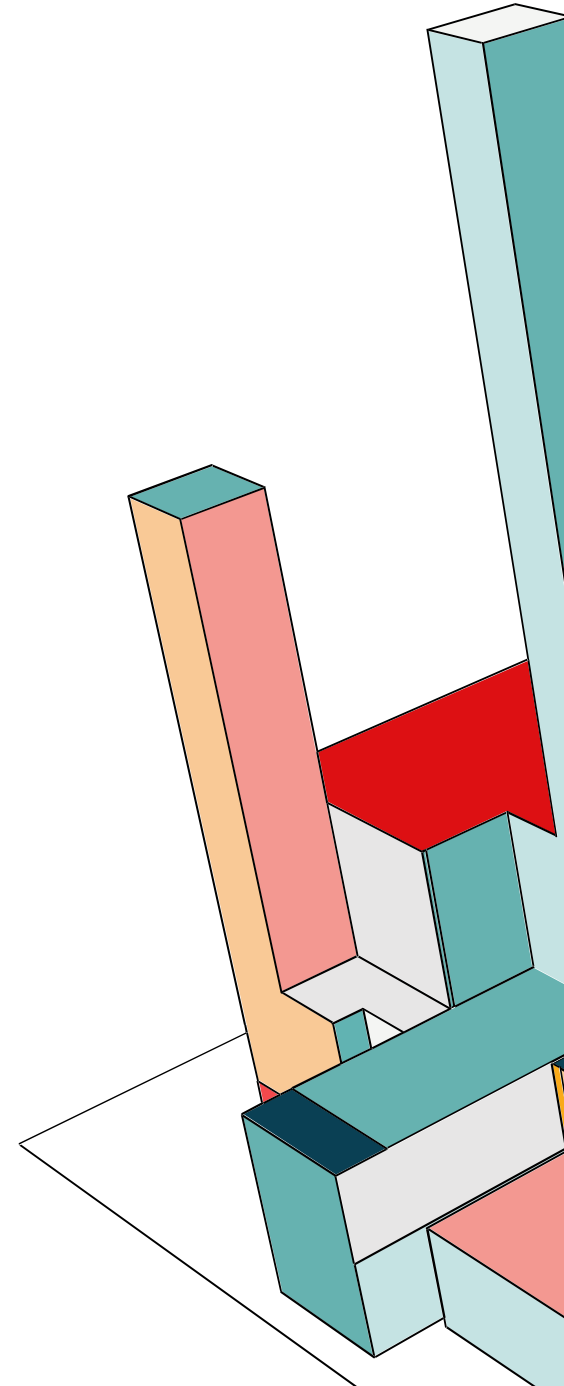




1. INTRODUCTION

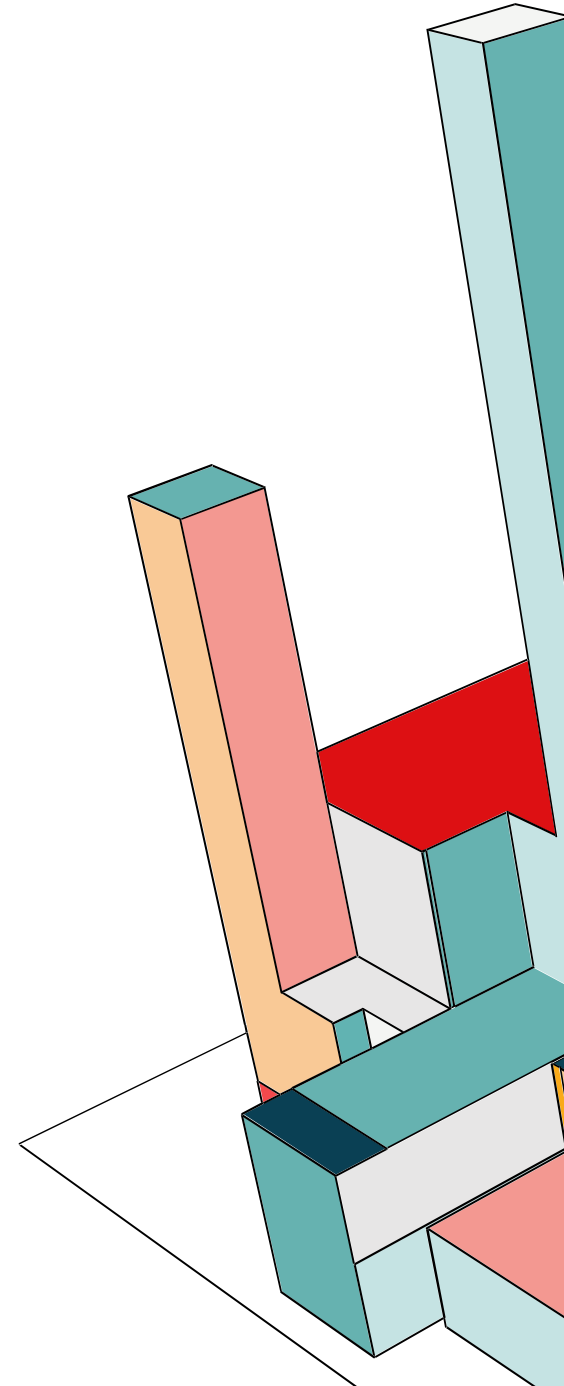
1.1. OBJECTIVES

The objectives of this assignment is confidence-building strategies to approach the varied- factors affected loan default at Home Credit – one of the famous consumer finance company.



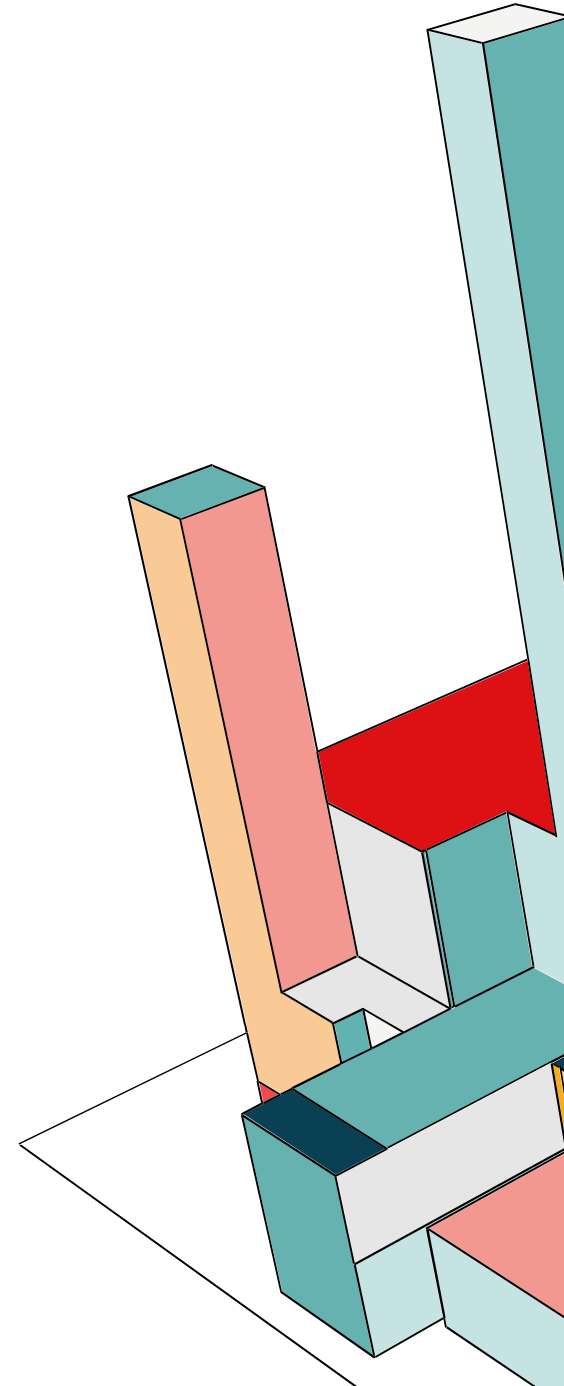
1.2. DATA USED

- Current Applications (application_data.csv): cur_app
- Previous Applications (previous_application.csv): pre_app
- Columns Dictionary (columns_description.csv): only referral for insights purpose



1.3. DATA OVERVIEW

- **Shape of the Datasets (Rows, Columns):**
 - Shape of the DataFrame Current App: (236175, 122)
 - Shape of the DataFrame Previous App: (1670214, 37)
- **General of Data Types:**
 - Number of columns for each data type in the **Current Application** dataset:
 - int64: 44
 - float64: 26
 - object: 20
 - category: 5
 - Number of columns for each data type in the **Previous Application** dataset:
 - int64: 6
 - float64: 15
 - object: 16

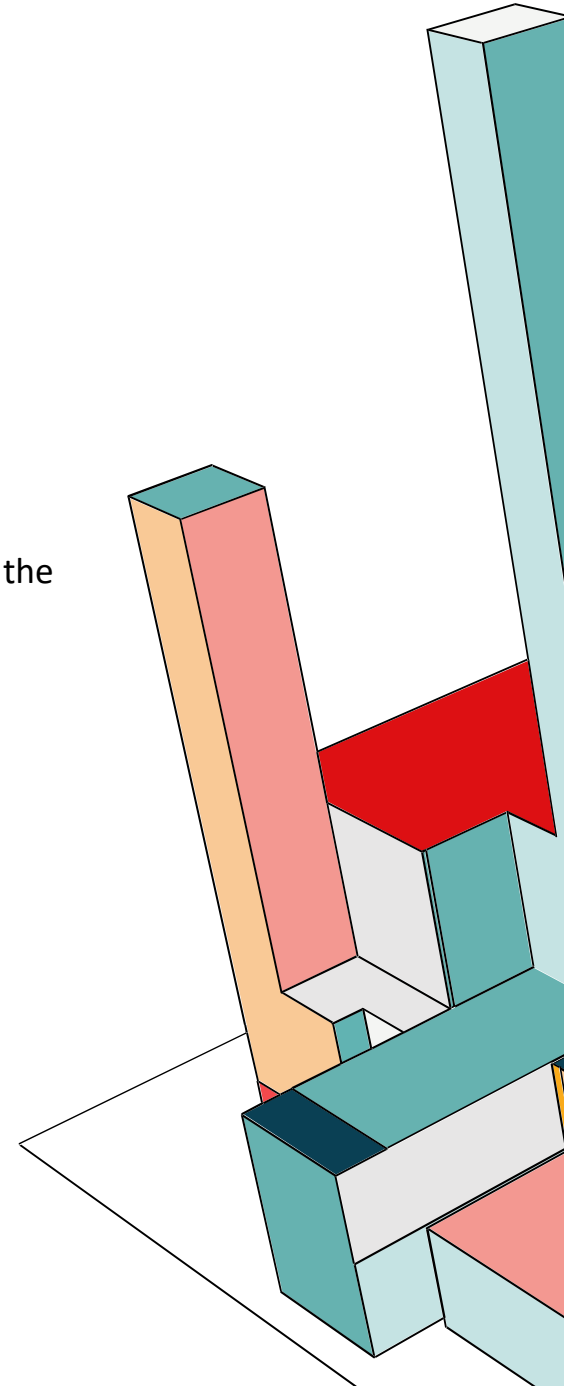


2. DATA QUALITY AND FEATURE PROCESSING



2.1. DATA QUALITY CHECK

- **Handling Missing Values:** Check columns with >30% missing values.
 - Fill null values by Other in OCCUPATION_TYPE column
 - Remove all suffixes columns (end with 'AVG', 'MODE', or 'MEDI') having >30% missing values
 - Consider remaining missing values, maintain reasons:
 - OWN_CAR_AGE (66%): Not all clients having car so missing value is reality
 - EXT_SOURCE (19% - 56%): This data is very important and define by other sources; by imputing or deleting it would create the problem with data integrity
 - AMT_REQ_CREDIT_BUREAU (13%): This data is provided by credit bureau and not all clients having the data their so keep it as it is
 - Other columns having small data missing so keep as it is.
- **Duplicate Description Checks:** Check FLAG_PHONE is the same FLAG_WORK_PHONE or not as description is the same
- **Feature Transformation:**
 - Converted DAYS columns (e.g., DAYS_BIRTH) to years (e.g., AGE, YEARS_EMPLOYED).
 - Check reasons value: -1000 of YEARS_EMPLOYED and convert to normal parameter
 - Reformat the NAME_EDUCATION_TYPE to make it better sorting
 - Check low unique value columns:
 - Create TOTAL_DOCUMENTS_PROVIDED by combining all FLAG_DOCUMENT columns
 - Create TOTAL_CONTACT_METHODS by combining all contact methods columns



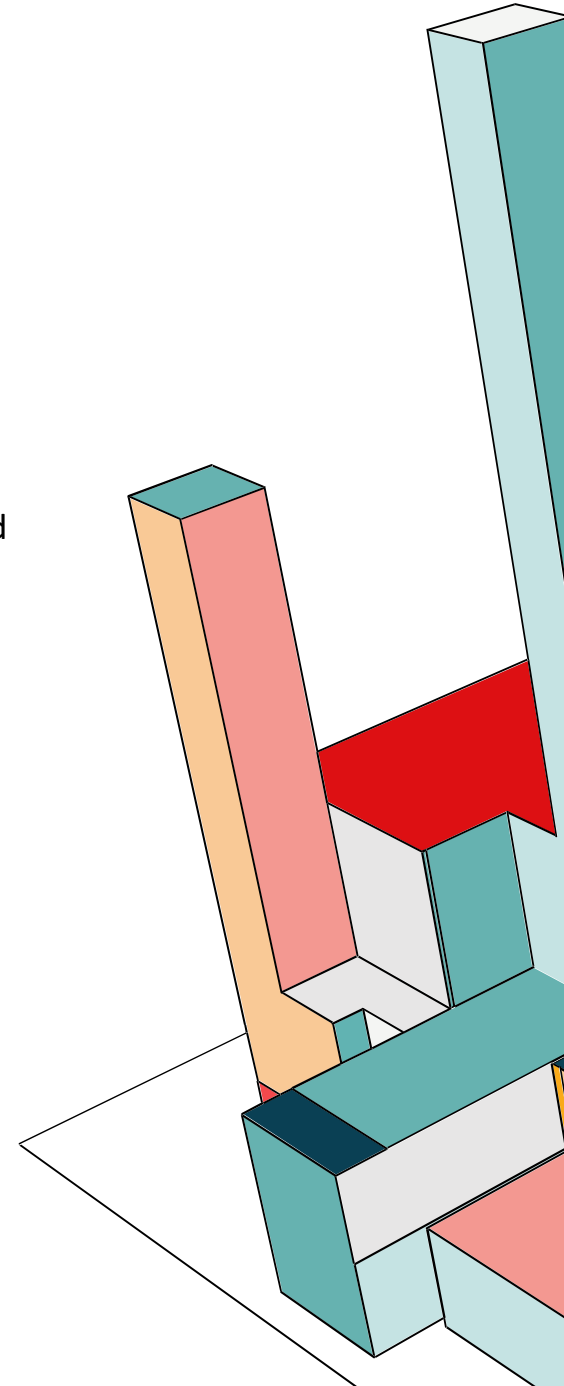
2.2. FEATURE ENGINEERING

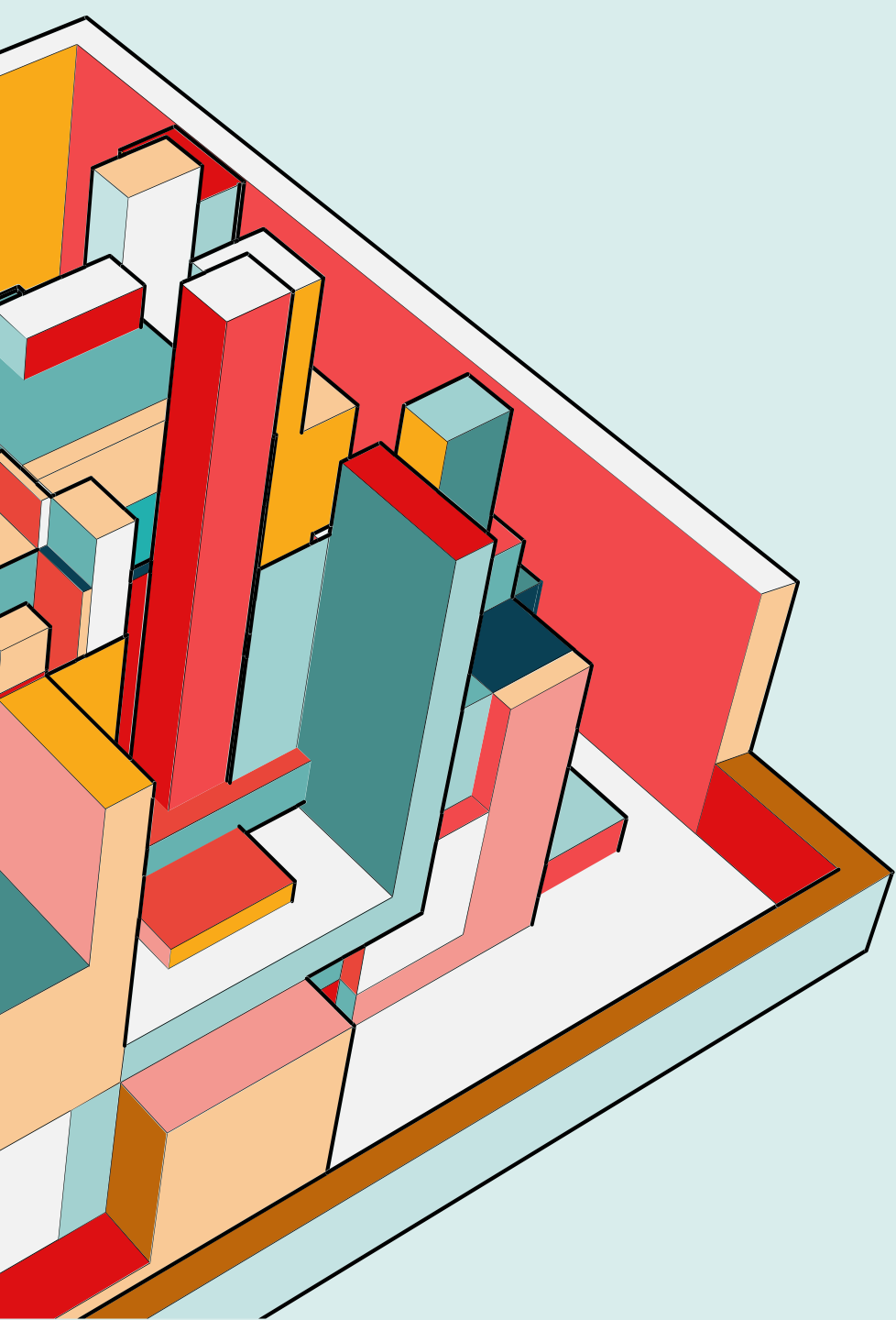
- **Group Categorization:**

- Children Group: Created CHILDREN_GROUP.
- Income Group: Created INCOME_GROUP.
- Years Registration Group: Created YEARS_REGISTRATION_GROUP.
- Years ID Publish Group: Created YEARS_ID_PUBLISH_GROUP.

- **New Ratio Calculation**

- Provided new ratio columns: CREDIT_INCOME_RATIO, MONTHLYCREDIT_INCOME_RATIO, and TENOR, and categorized them.





3. ANALYSIS OF PREVIOUS APPLICATIONS

3. ANALYSIS OF PREVIOUS APPLICATIONS

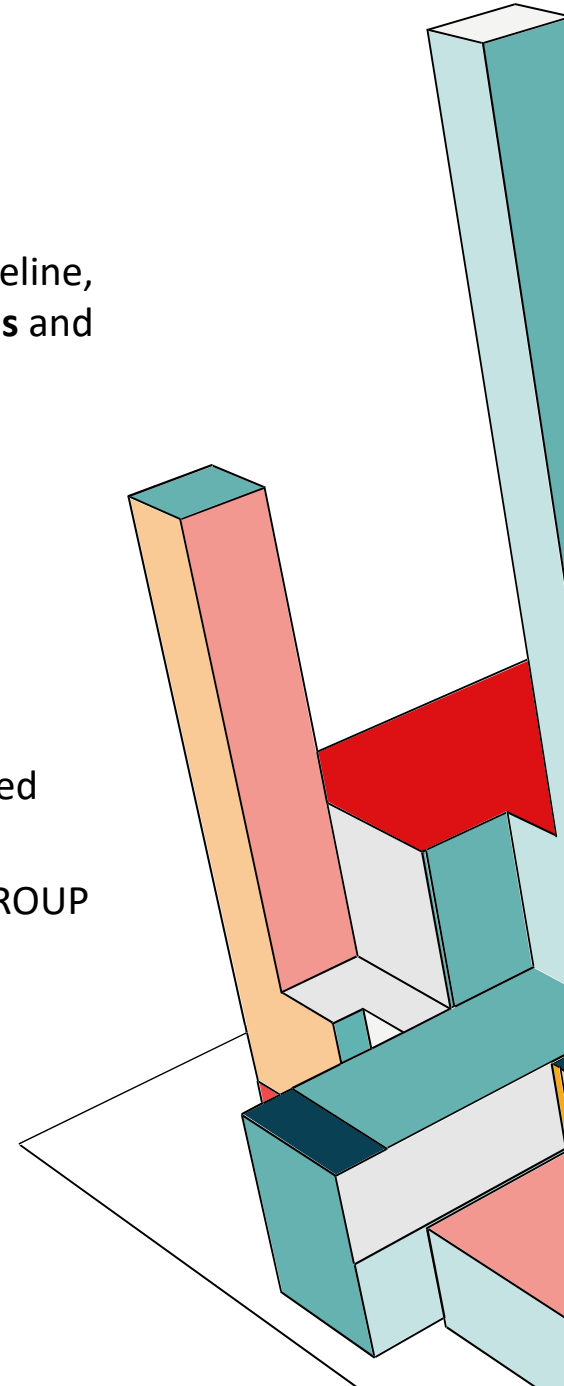
The data in previous applications have a lot of rooms to do the analysis but due to limitation in timeline, taken the important feature in the previous applications by tracing backs **how long clients are with us** and **how many applications they are with us** to see the impact with default rate

- **Approved Application Count:**

- Take only 'Approved' in NAME_CONTRACT_STATUS to filter previous contract have approved
- Calculated NUM_APPROVED_PREVIOUS_APPLICATIONS.
- Categorize numbers of Approved Application with column PREAPP_GROUP

- **Time Since Last Approved Application:**

- Created YEARS_SINCE_PREV_DECISION to calculate the time difference since the last approved application.
- Categorize numbers of Approved Application with column YEARS_SINCE_PREV_DECISION_GROUP



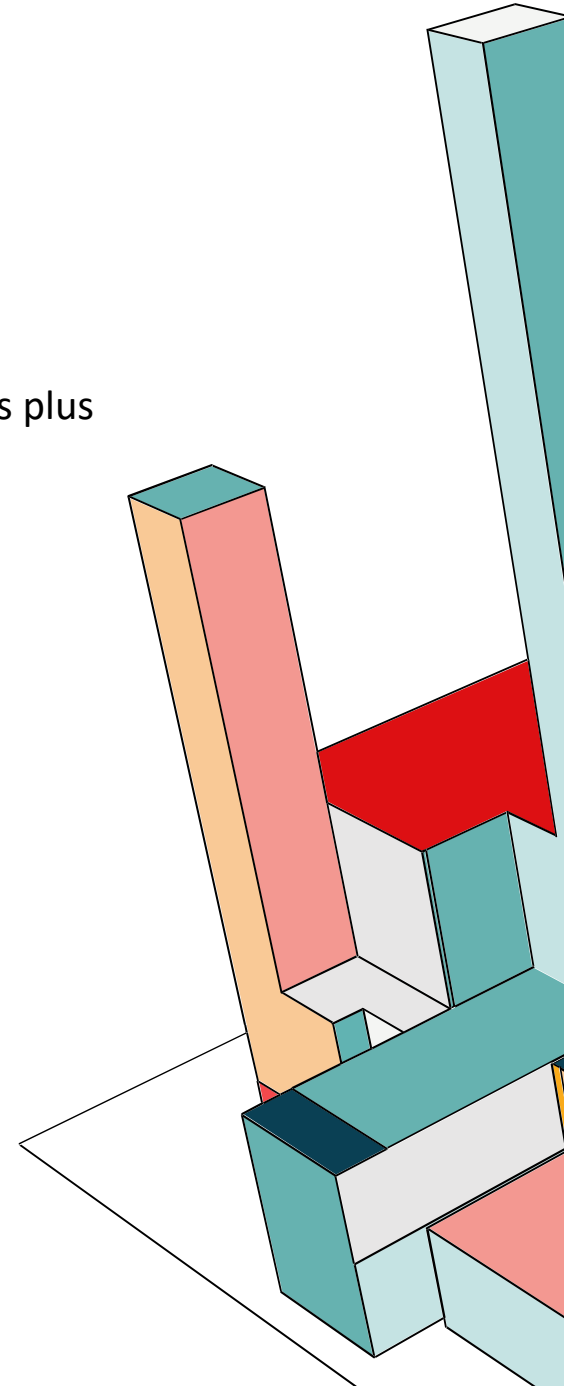
4. NUMERICAL UNIVARIATE ANALYSIS



4. NUMERICAL UNIVARIATE ANALYSIS

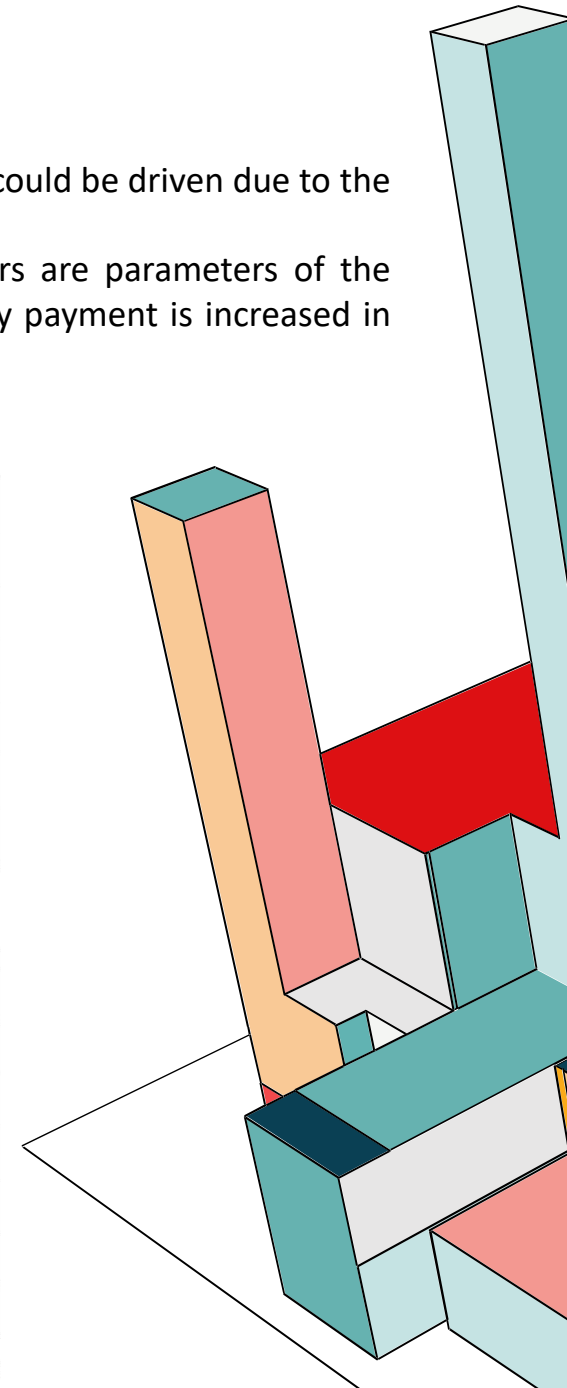
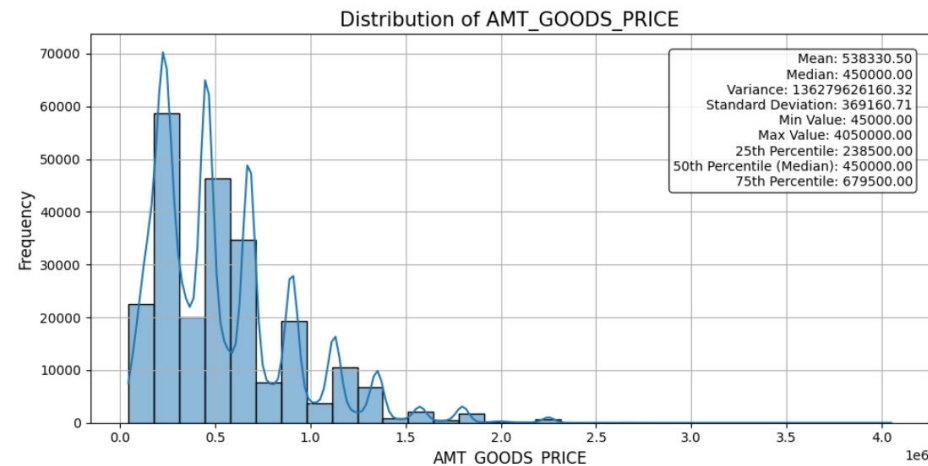
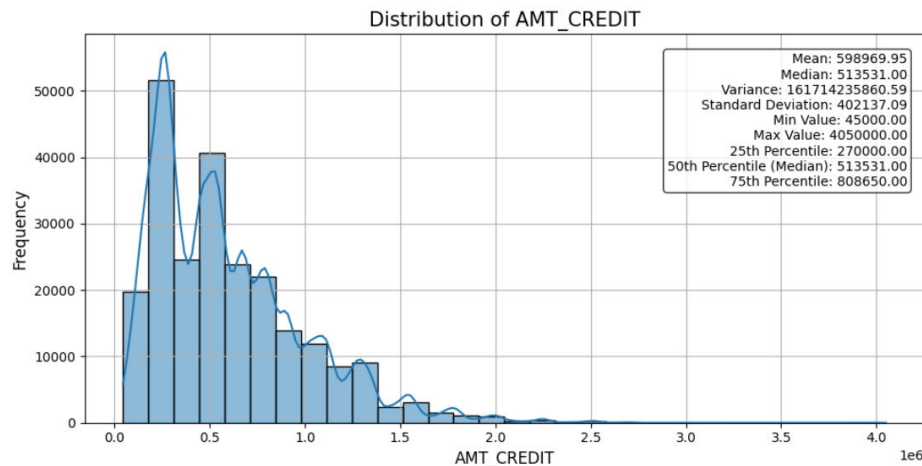
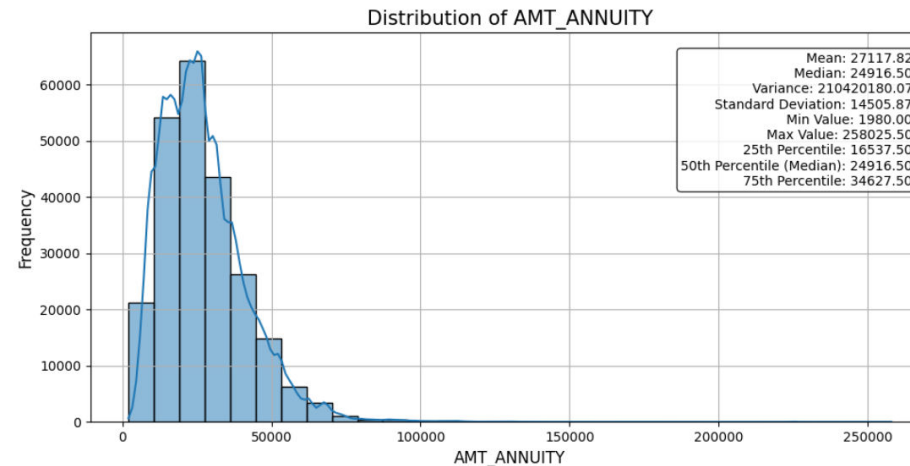
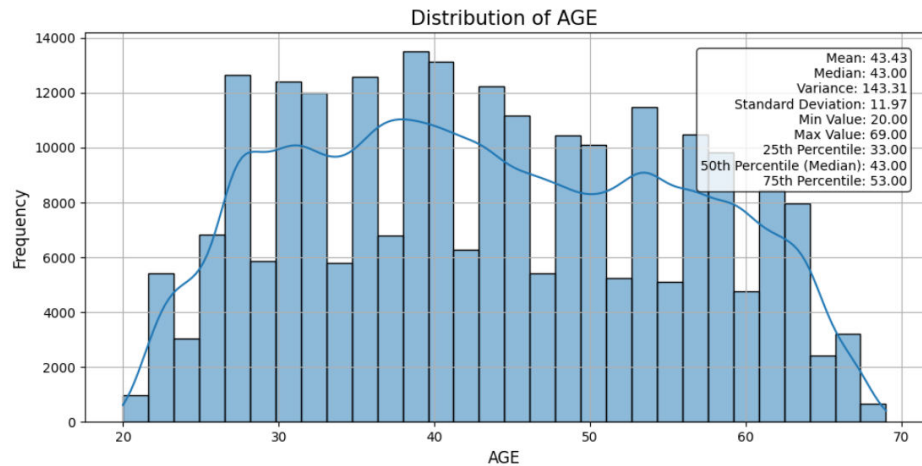
We would take the numerical columns not the binary columns also columns have small unique values plus the key ID (SK_ID_CURR)

- **Overview:** Analyzed distributions of selected numerical columns excluding binary features.
- **Visuals:** Histograms showing key statistics for numerical features
- **Statistics on Plots:** Included mean, median, variance, standard deviation, minimum, and maximum values for each feature.



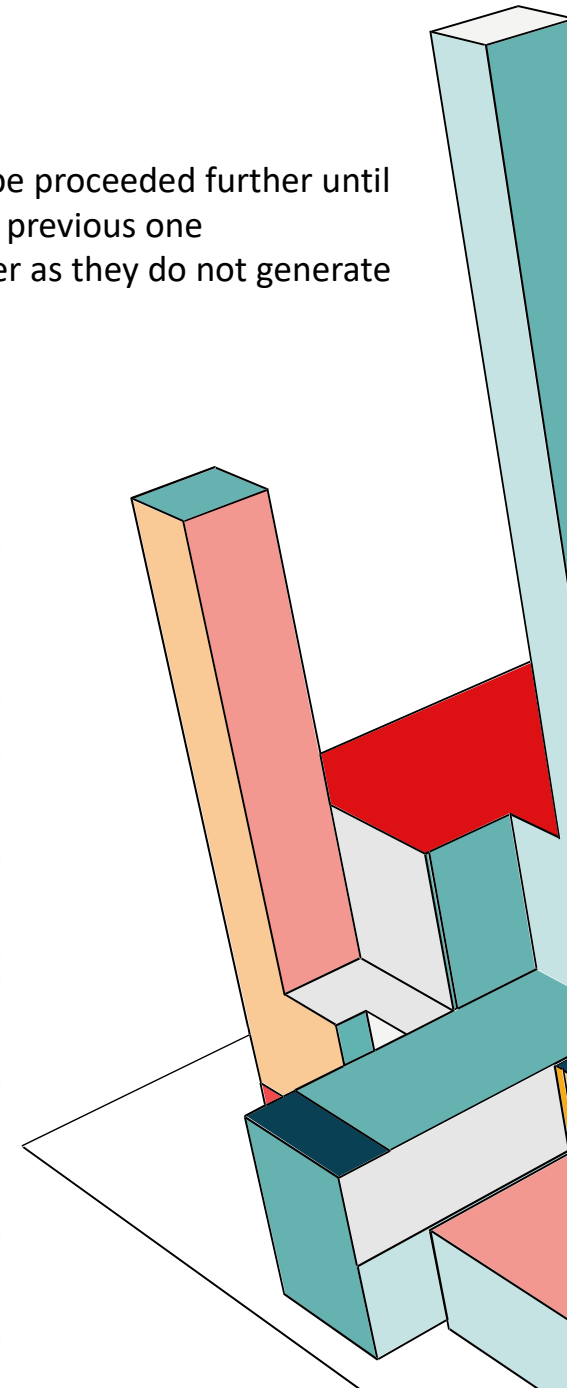
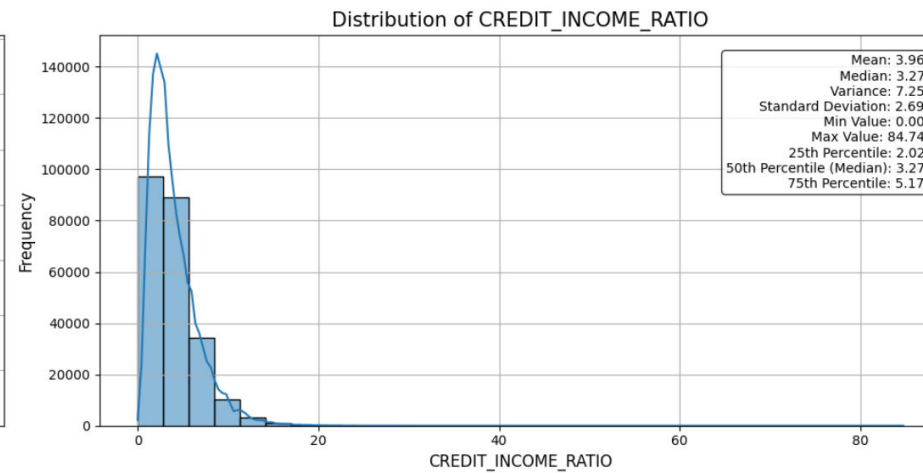
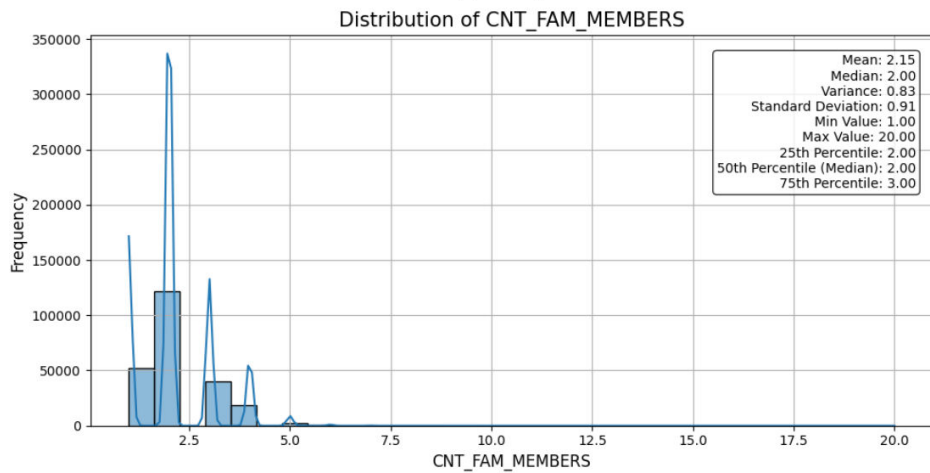
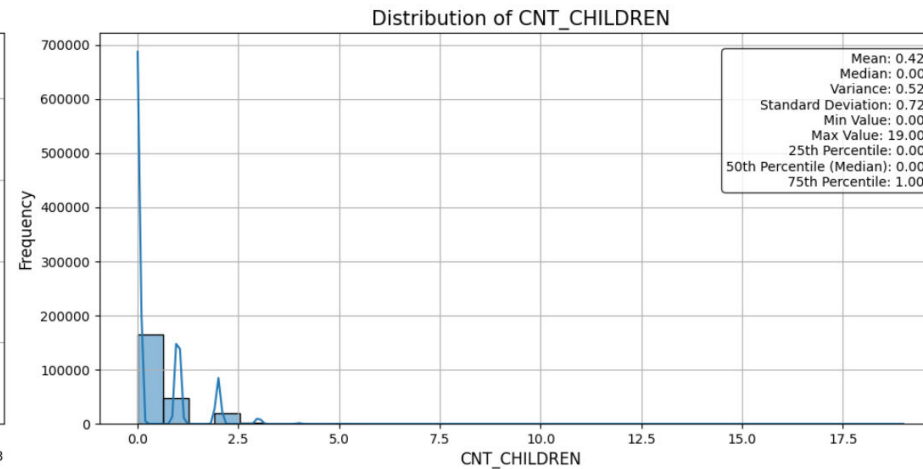
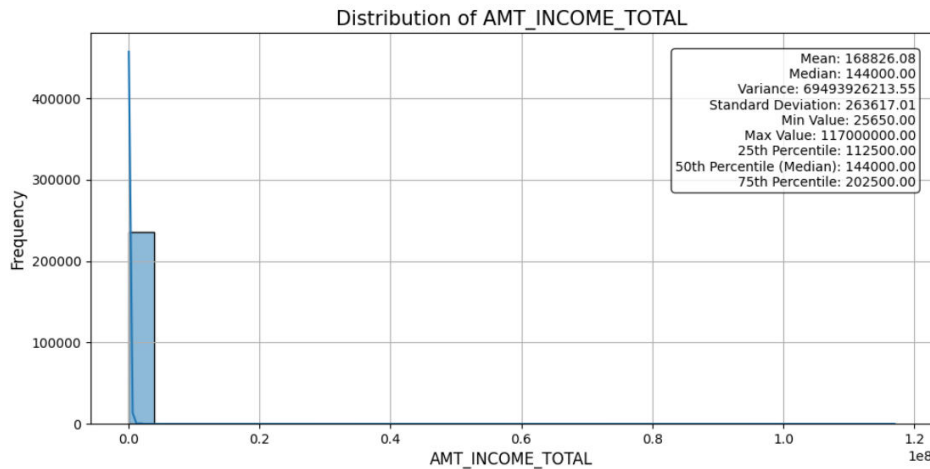
4. NUMERICAL UNIVARIATE ANALYSIS

- Distribution of age is evenly, and the minimum age is requested from 20 but it is not too big compared >25. It could be driven due to the risk in that age group is big
- Amount of Annuity, Credit Amount and Goods Price is distributed the same trend as those kind of factors are parameters of the application. Usually, to buy higher goods price they must take bigger credit and because of that the monthly payment is increased in parallel



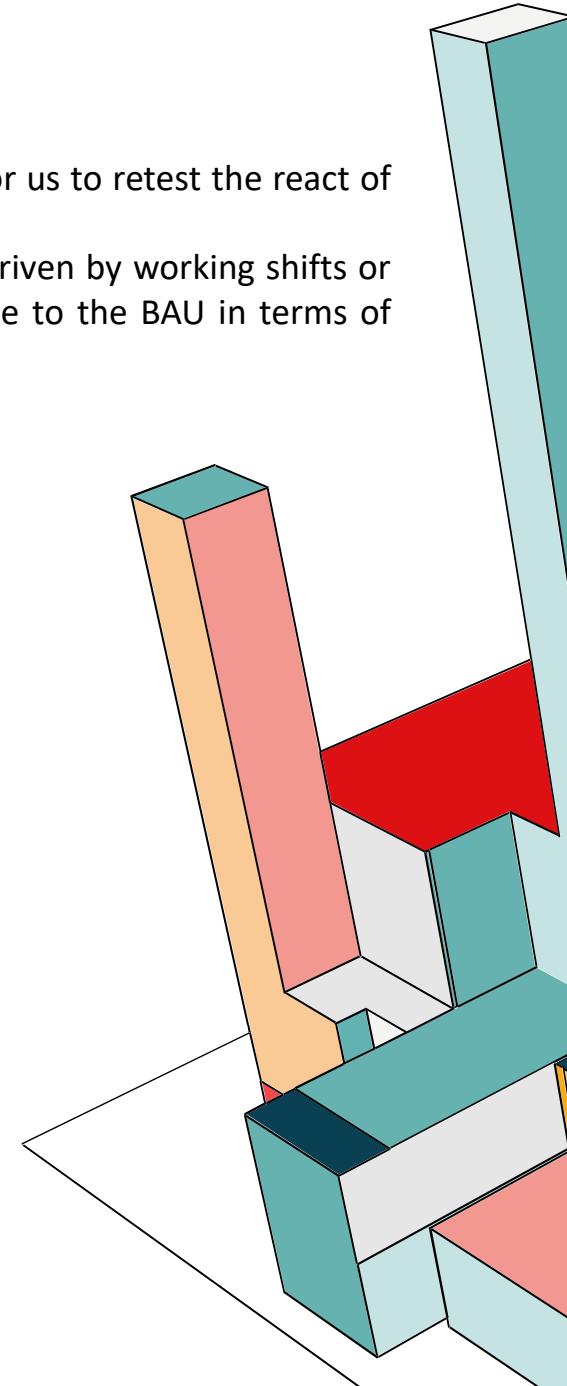
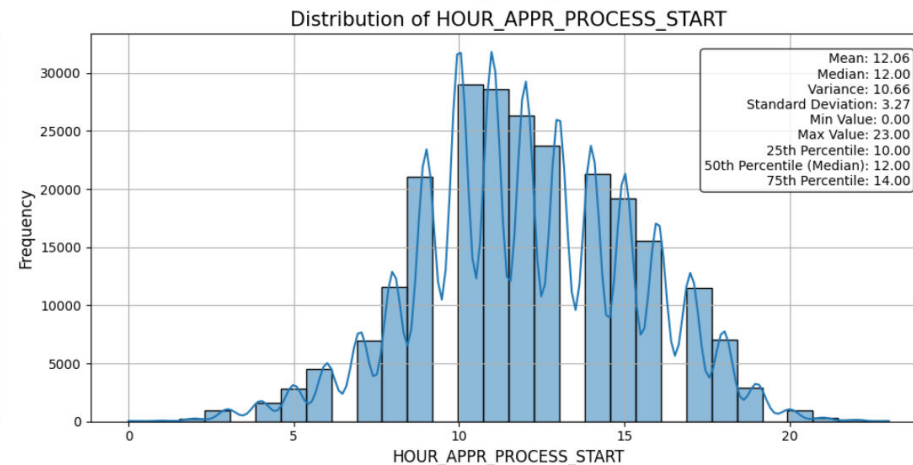
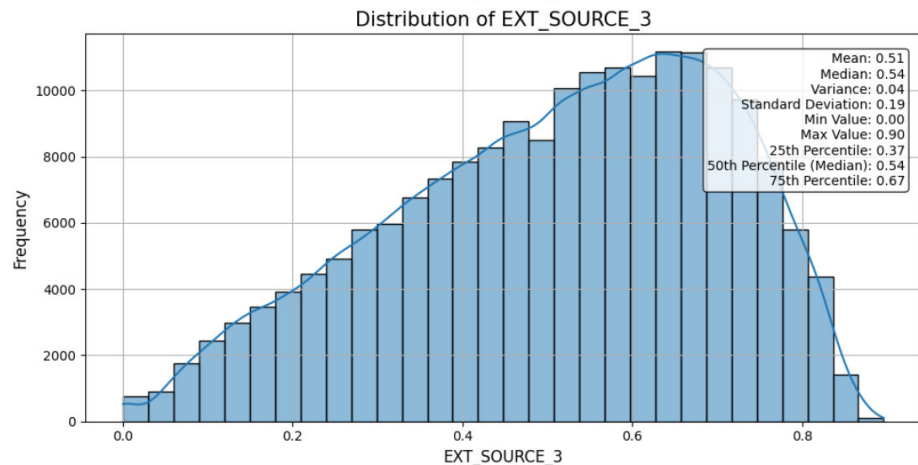
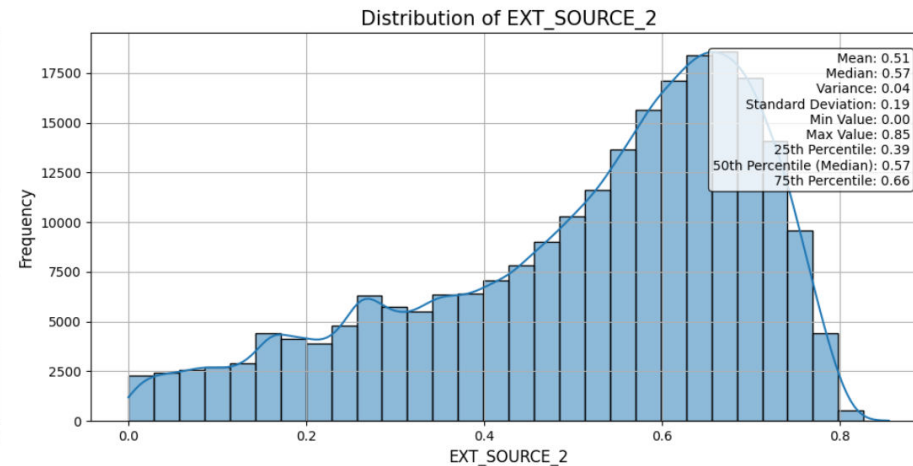
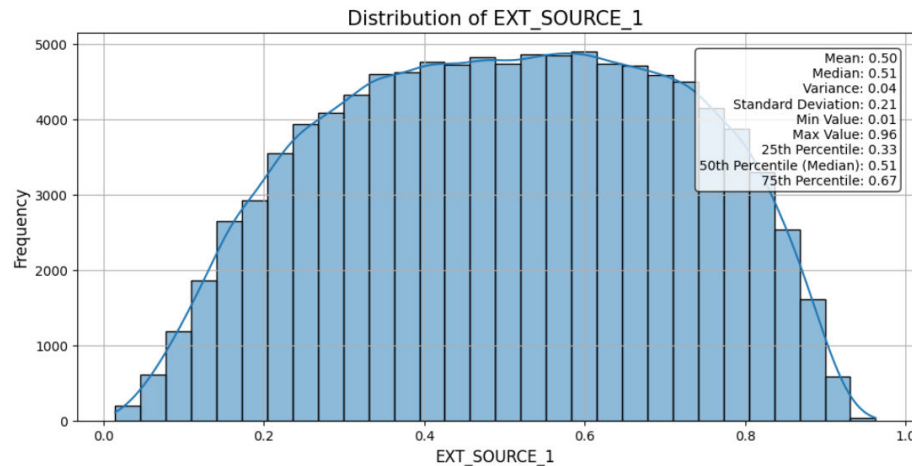
4. NUMERICAL UNIVARIATE ANALYSIS

- Data in AMT_INCOME_TOTAL are totally imbalanced as all of frequency shifting to the left; those data may not be proceeded further until it changes to ratio. Then, when looks to CREDIT_INCOME_RATIO, the distribution looks better compared to the previous one
- Number of family members and numbers of children is imbalanced but to move on number of children is matter as they do not generate income, and it is more valued compared to family members



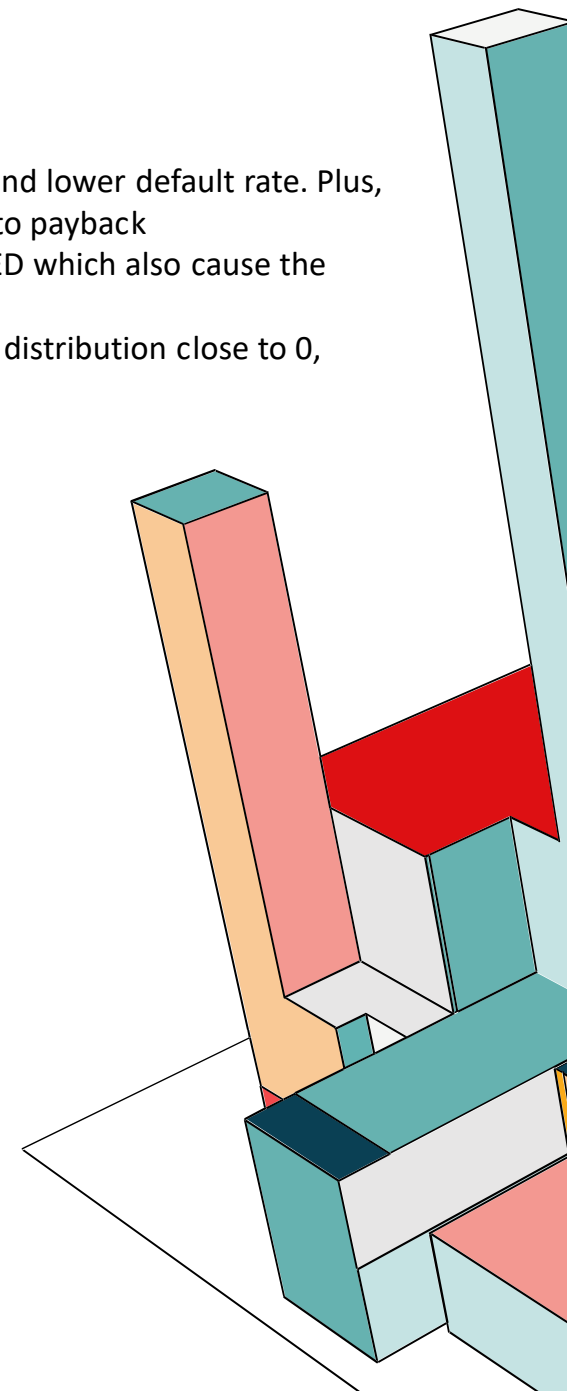
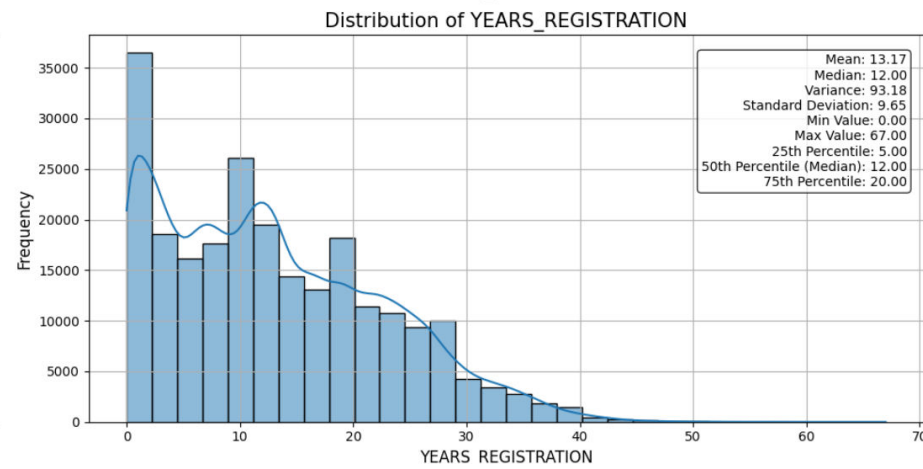
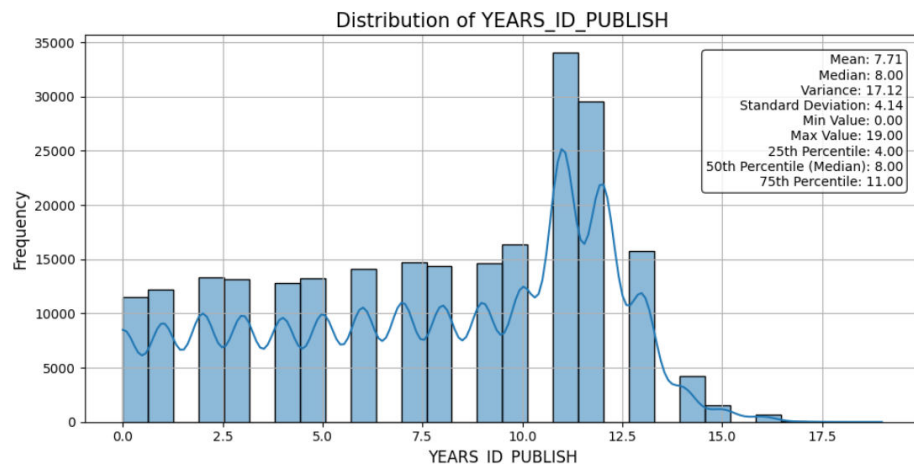
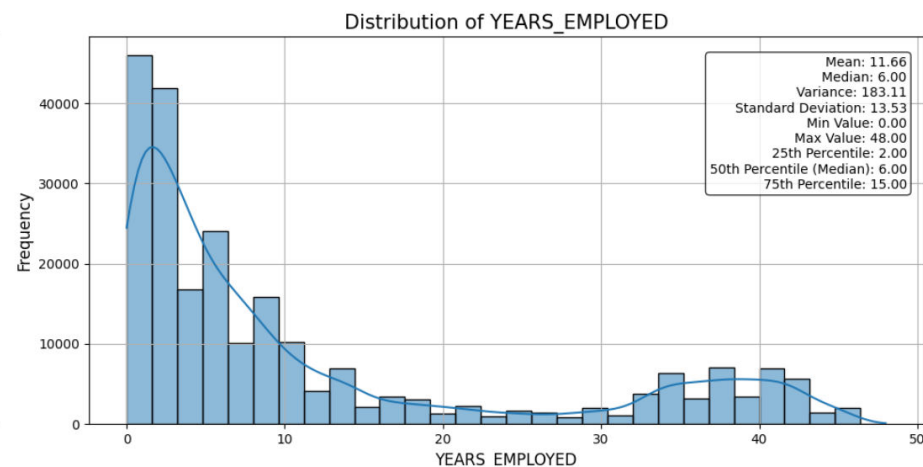
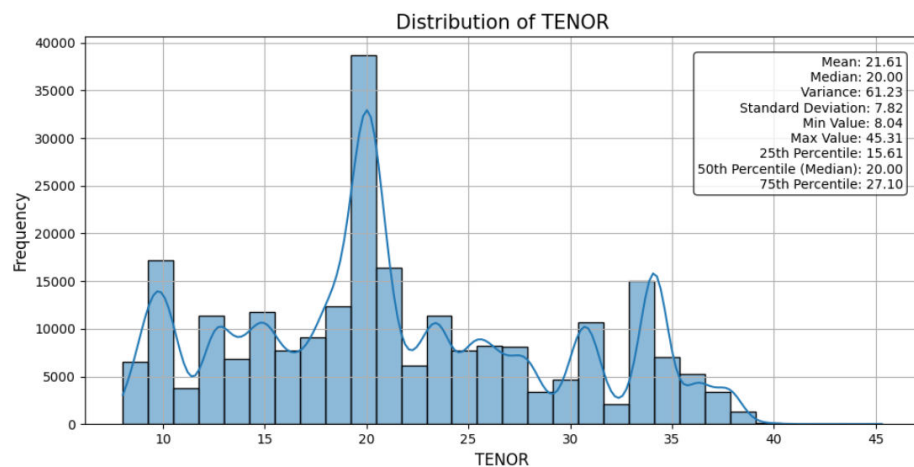
4. NUMERICAL UNIVARIATE ANALYSIS

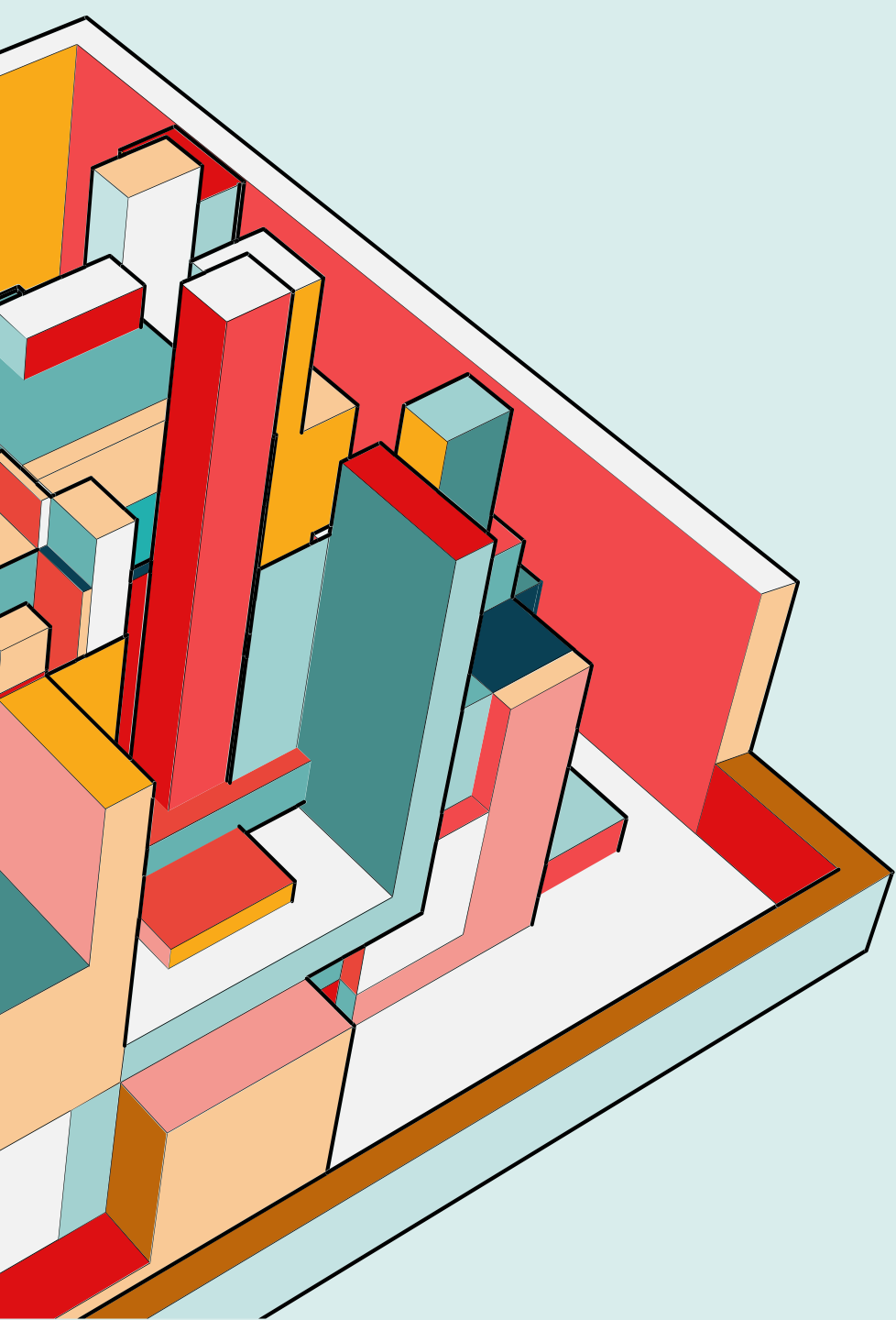
- Data of EXT_SOURCE is not ready at this stage but somehow it is distributed evenly, and it would be useful for us to retest the react of the data to default loan.
- The HOUR_APPR_PROCESS_START has shown the bins are not stable there is the drop in the day, it can be driven by working shifts or system not stable, however those kind of the data has not proved much in the business of default loan due to the BAU in terms of processing time



4. NUMERICAL UNIVARIATE ANALYSIS

- Data of TENOR is created to observe the terms of applications and usually tenor longer would be benefit for company and lower default rate. Plus, the shorten tenor from 0-12 maybe consider as they are also having needy the credit in short term, and they are likely to payback
- YEARS_EMPLOYED is very important and the distribution prove that there are many clients having low YEARS_EMPLOYED which also cause the default rate is higher while the group of higher YEARS_EMPLOYED will have better default rate
- YEARS_ID_PUBLISH and YEARS_REGISTRATION also a good factor to monitor the performance of default rate and when distribution close to 0, there are more likely that default rate would increase as the frequency of changes is high





5. CATEGORICAL AND SEGMENTED UNIVARIATE ANALYSIS

5. CATEGORICAL AND SEGMENTED UNIVARIATE ANALYSIS

Overview: As the time limit, take out all the categorical columns and evaluate the single impact of each factor to the default rate by Data Frame with those values:

- Name of Column
- Gap between lowest default rate value and highest default rate value
- Find out how many unique values the factor having
- Highest Default Rate Value
- Lowest Default Rate Value

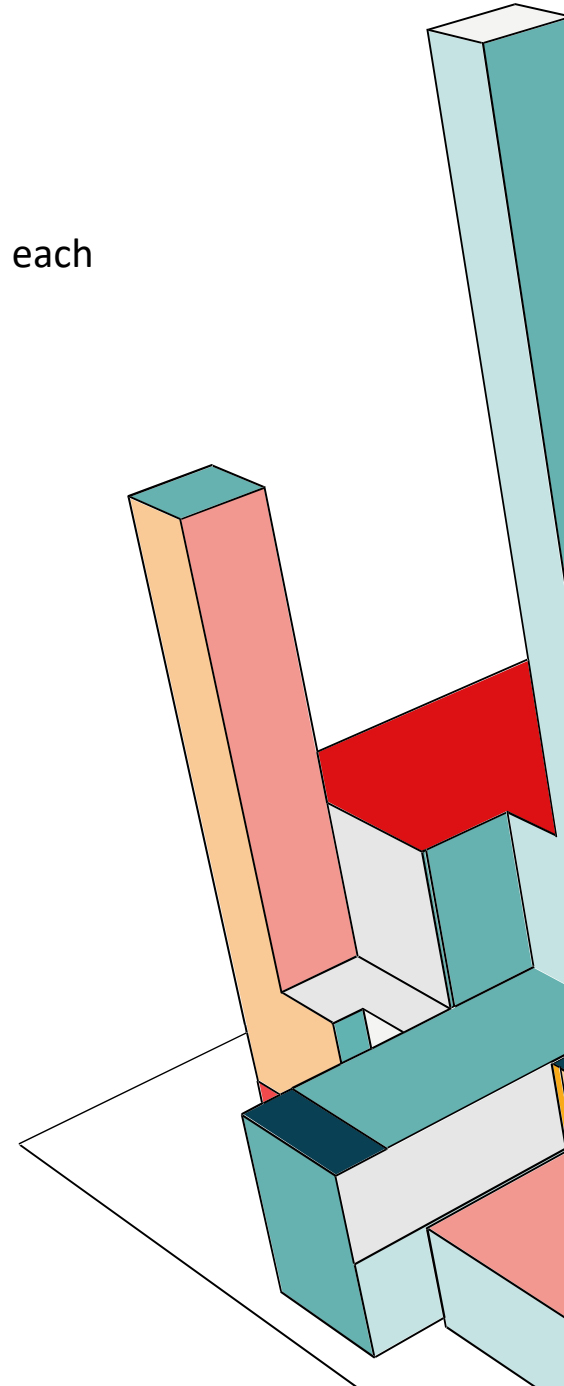
Chose 10 biggest gap factors for further analysis

Visuals: Limit the number of unique value by grouping them to easier for EDA.

- Using stacked bar charts for checking defaulted rate
- Using bar charts to see the frequency of each item in the factor

Evaluations:

- Check the categorical ordered variables to see the trend of default rate among group
- Provide insights based on the stacked bar charts and business scope



5. CATEGORICAL AND SEGMENTED UNIVARIATE ANALYSIS

OCCUPATION_TYPE Column

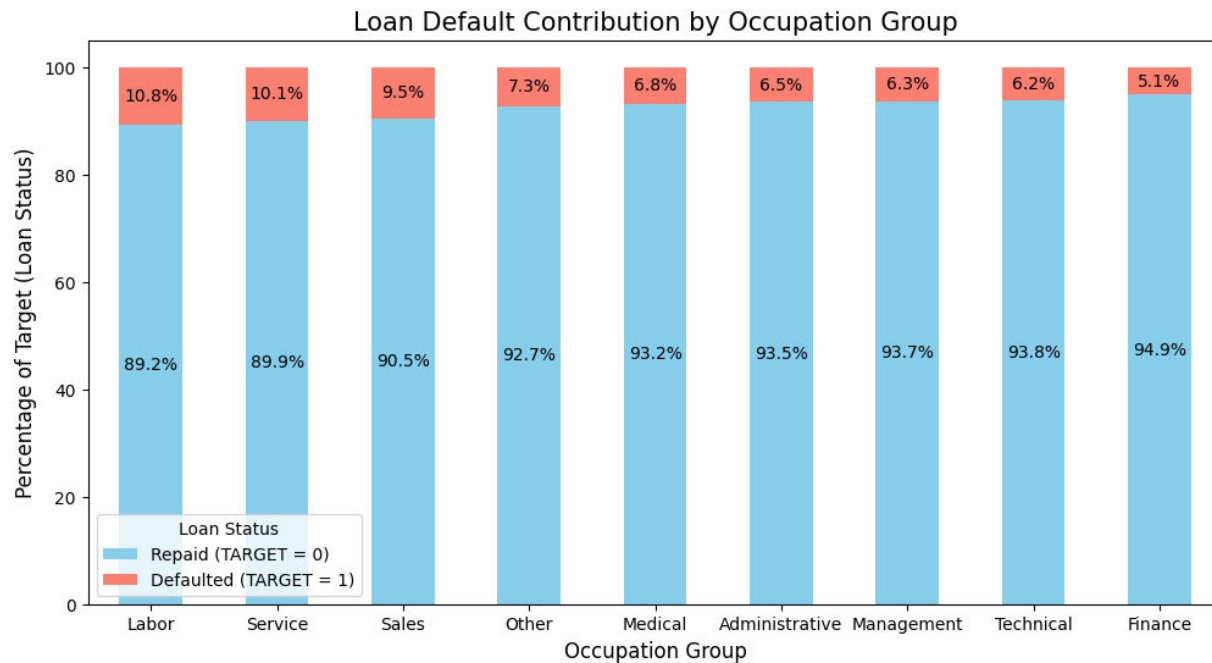
Problem: There are 58 unique values, and it is hard to make the analysis with the bar chart

Solution:

- Step 1: Group all the items having ‘%: type%’
- Step 2: Mapping the OCCUPATION_TYPE into broader group from 35 unique values to 9 unique values

Insights:

- Clients working on segment regarding Labor, Service and Sales having high default rate compared to remaining
- **Administrative, Management, Technical and Finance usually having better academic profile** and understanding, it could be found by the **matrix of these 9 groups with the education type** about the risk and the proactive to repay the loan

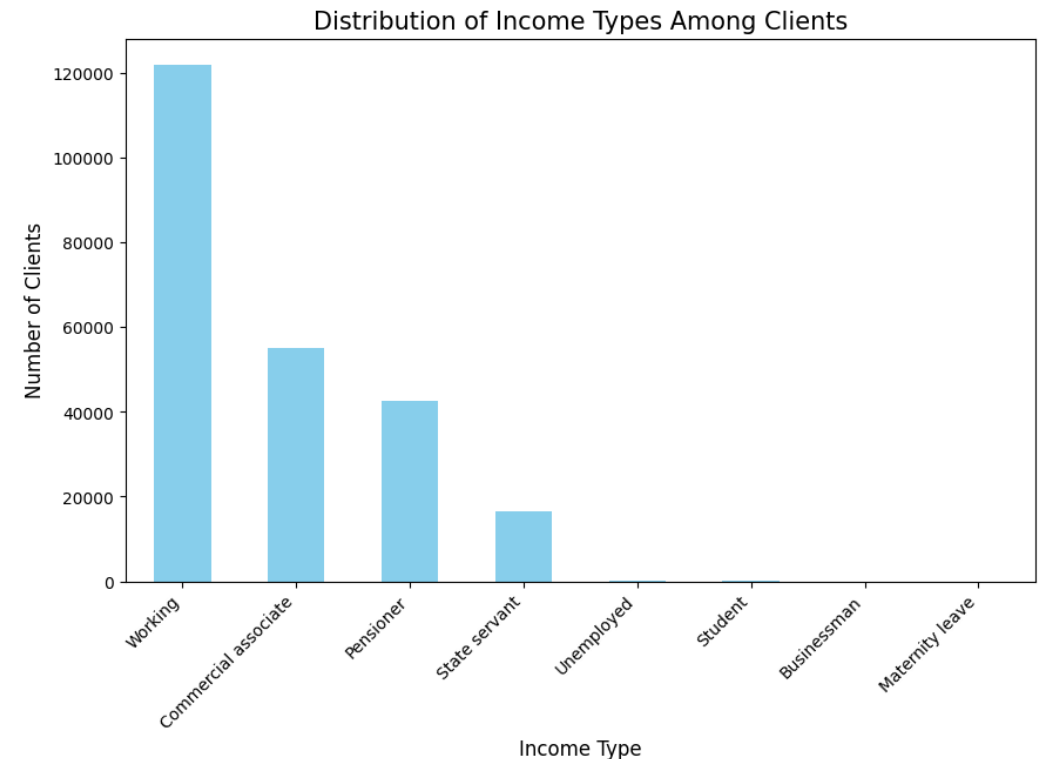
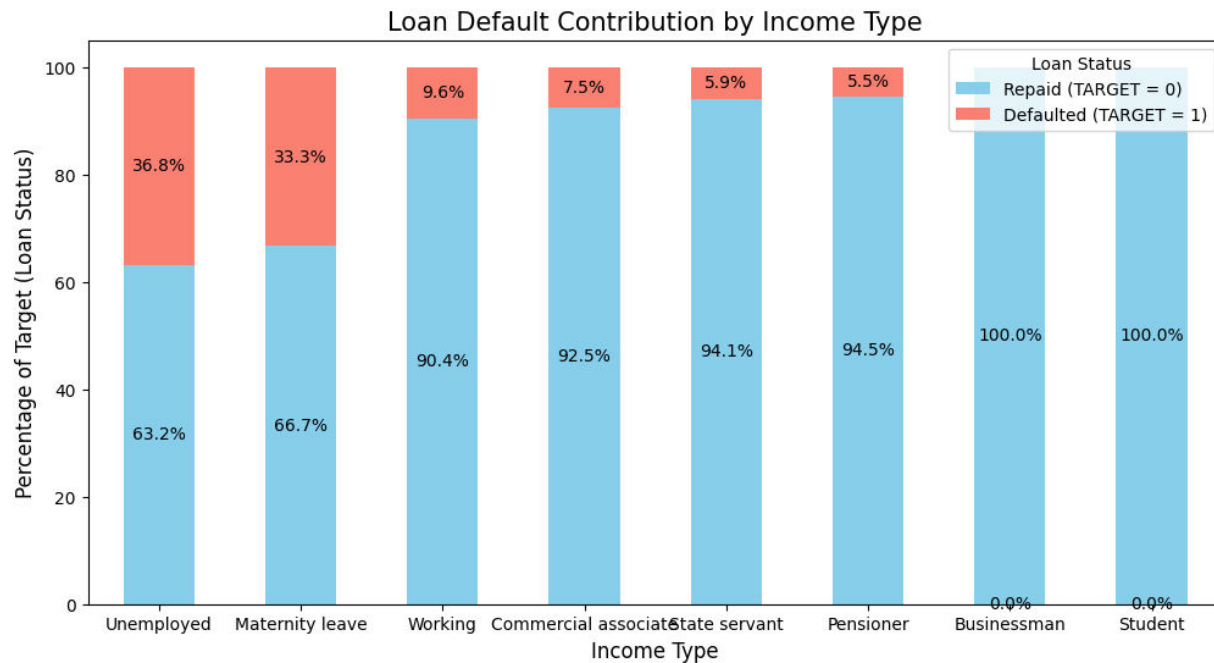


5. CATEGORICAL AND SEGMENTED UNIVARIATE ANALYSIS

NAME_INCOME_TYPE Column

Insights

- The Unemployed and Maternity Leave show significant of Default Rate compared to another group. However, the share of those group (right chart) has shown that there is significant low shares which can drive the unbalancing. But it would be the useful data to make the decision base on the Income Type for those client group
- The Working item also show the default rate higher compared to remaining group. However, the share of the clients in this group is significant compared to another group. **Suggestions: the data input should be more detailed in this group to make the good decision later**
- Pensioner also drive a good and stable result among the group which can be further analyze in Bivariate Analysis

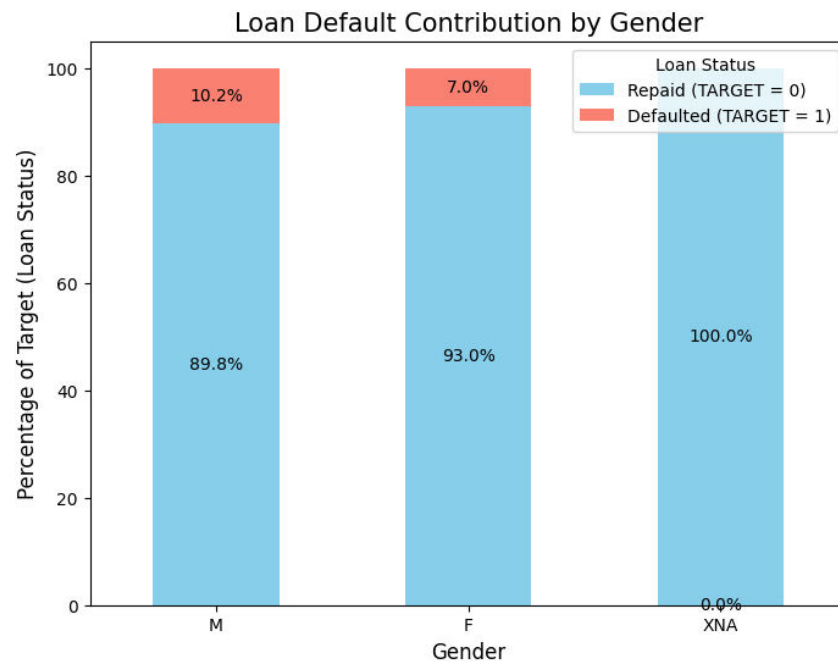


5. CATEGORICAL AND SEGMENTED UNIVARIATE ANALYSIS

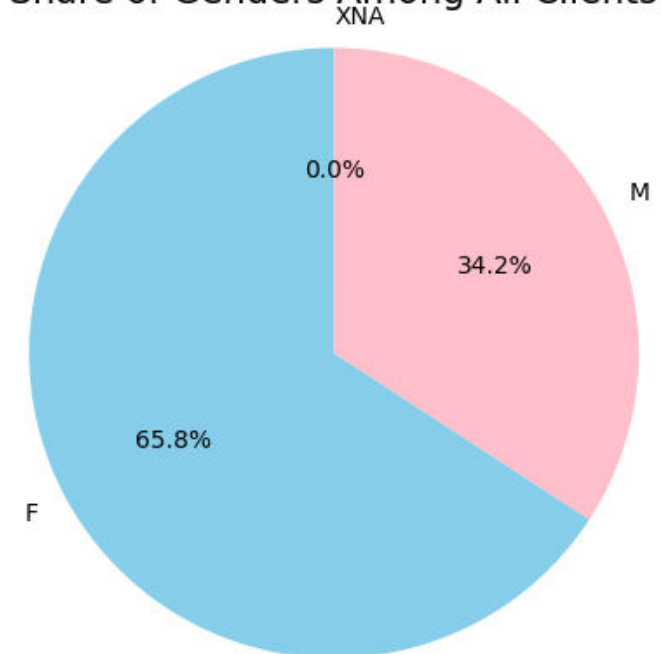
CODE_GENDER Column

Insights

- There are 4 clients cannot detect Gender but due to the sample is small and it would not impact the data overall so kept it as it is
- Males (M) always show the default rate higher than Females (F); we may put in Bivariate Analysis to see the heatmap with other factor (such as Age Group) to check whether Males' default rate always higher than Females' default rate



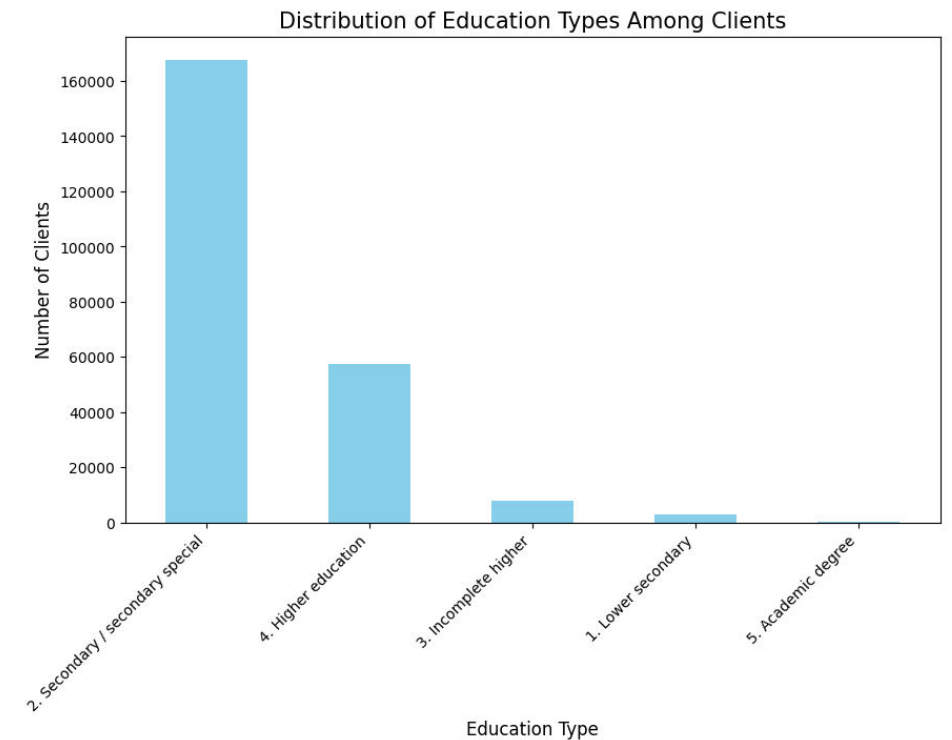
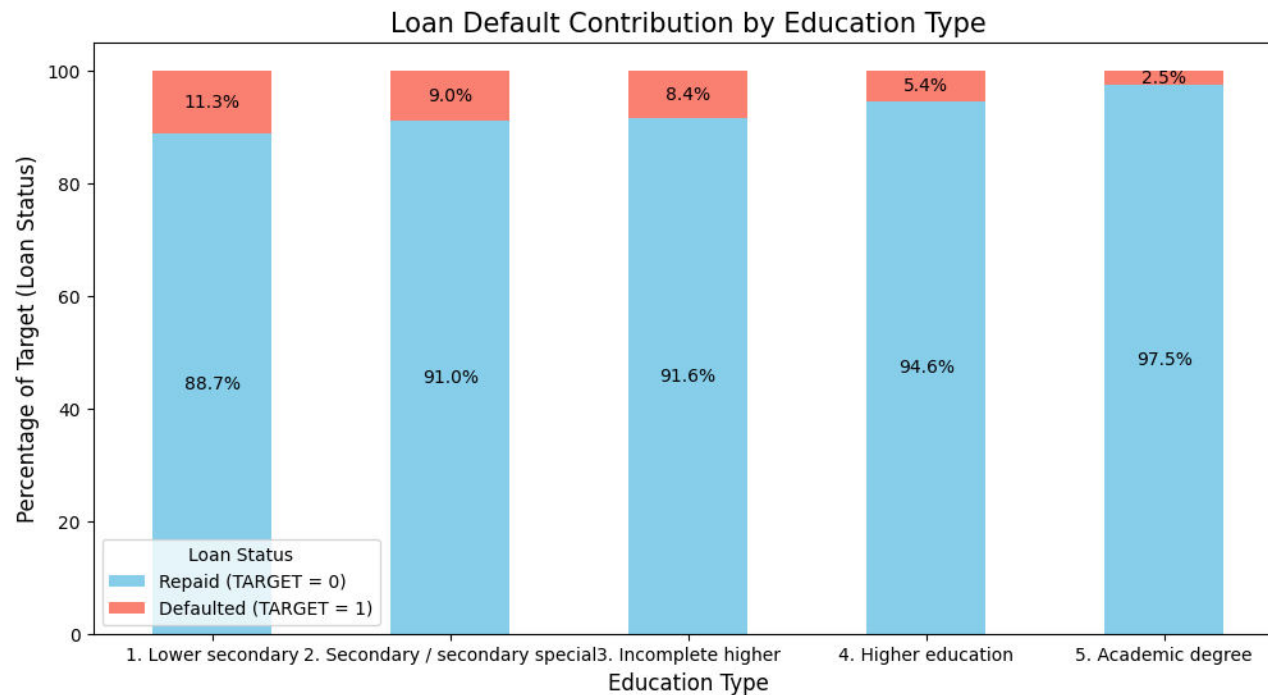
Share of Genders Among All Clients



5. CATEGORICAL AND SEGMENTED UNIVARIATE ANALYSIS

NAME_EDUCATION_TYPE Column

Insight: The more clients having higher education type the more understanding and proactive to prepaid the loan

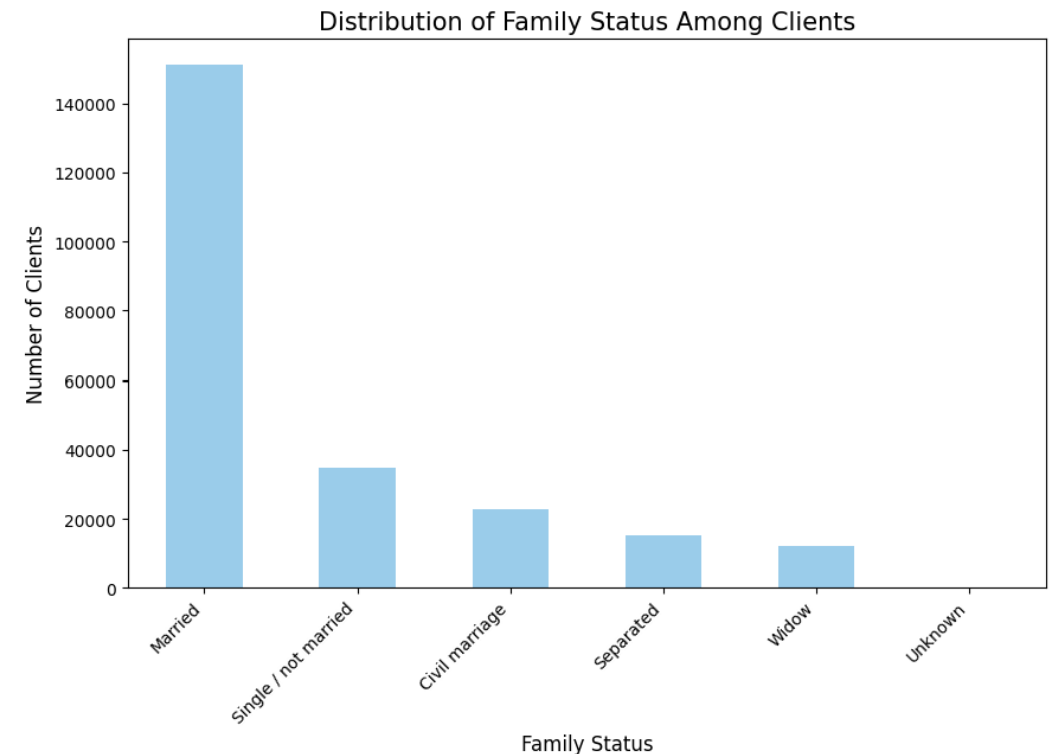
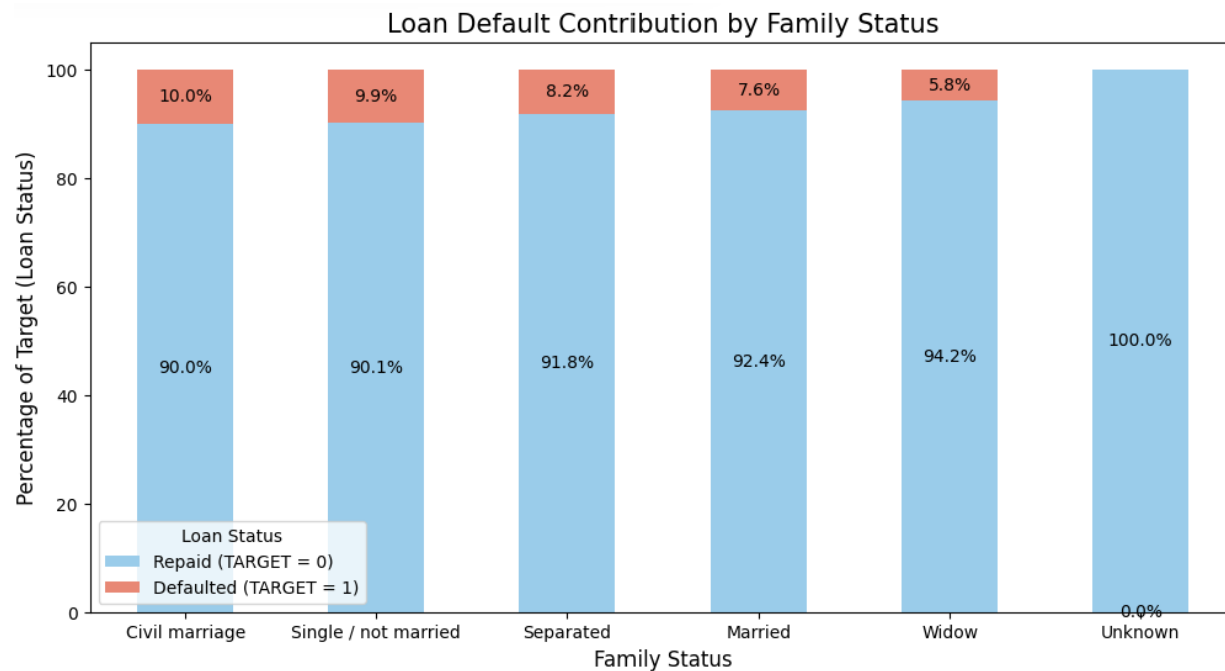


5. CATEGORICAL AND SEGMENTED UNIVARIATE ANALYSIS

NAME_FAMILY_STATUS Column

Insights

- The share of 'Civil marriage' - specifically refers to a marriage performed by a government authority and is generally non-religious and 'Single / not married' having significant default rate compared to other family status group.
- The surprised findings show that Widow having the lowest default rate which also a good insight for making decision in loan for Home Credit with this group of family status.

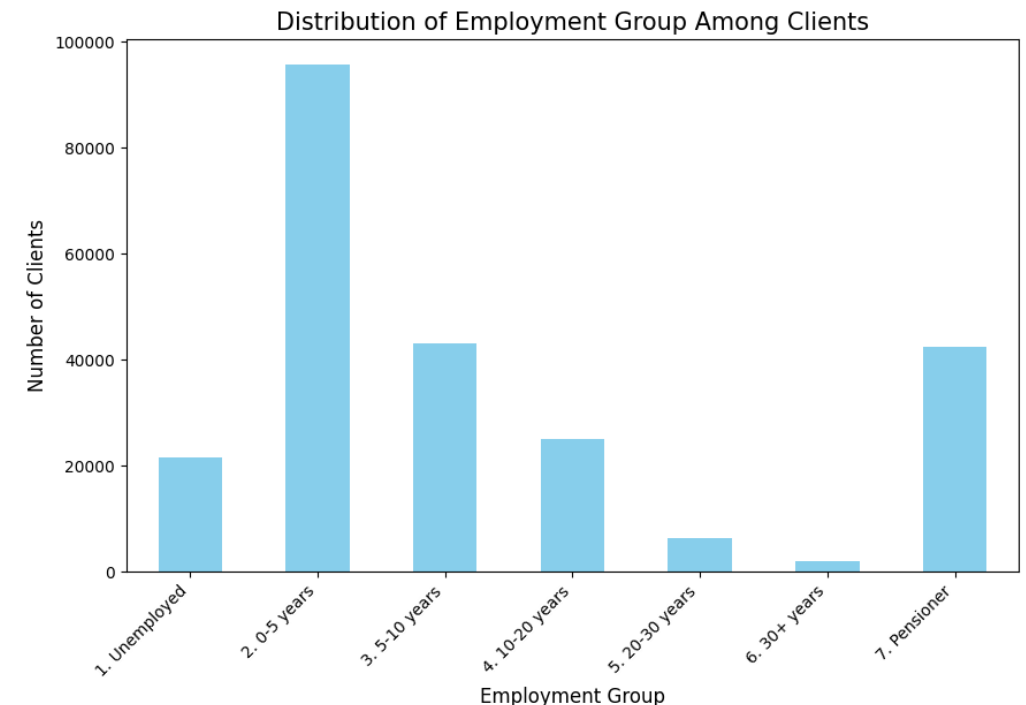
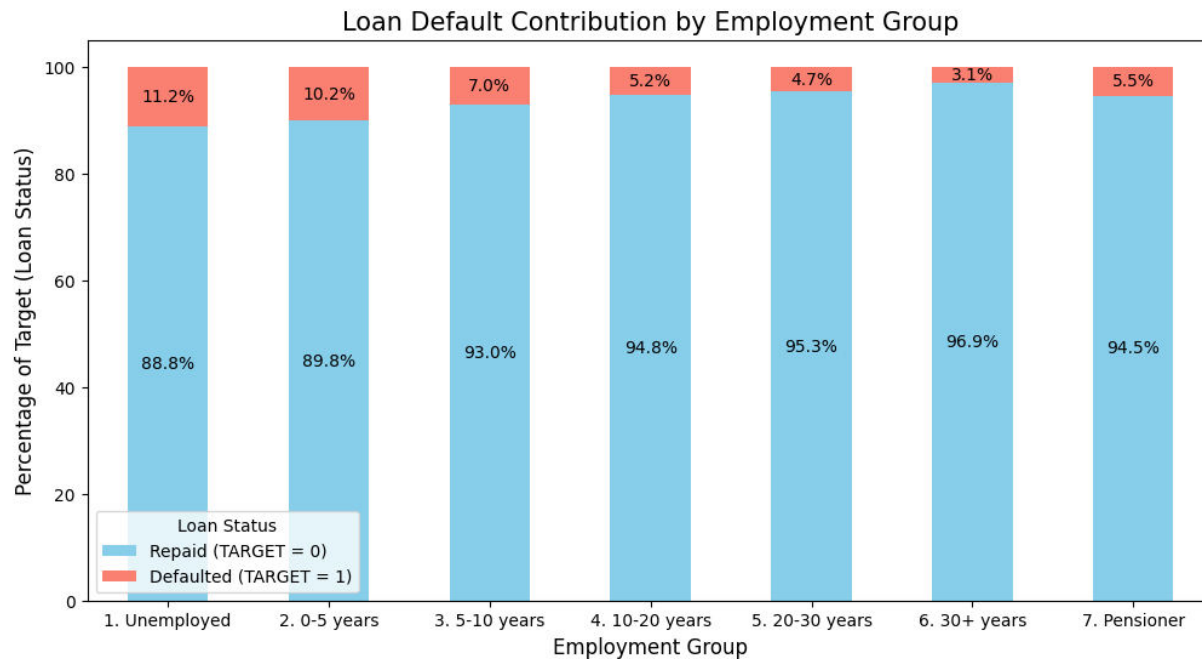


5. CATEGORICAL AND SEGMENTED UNIVARIATE ANALYSIS

EMPLOYMENT_GROUP Column

Insights

- The Unemployed group show the highest default rate, however, when checking back the name income type, the default rate among is different. Means, the data client's input may need further clarification in which Unemployed in the Employment Group but they still have other source of income type.
- The correlation between default rate and the years of employment group is parallel. The longer they worked the better default rate, especially with Pensioner which have the default rate at suitable level

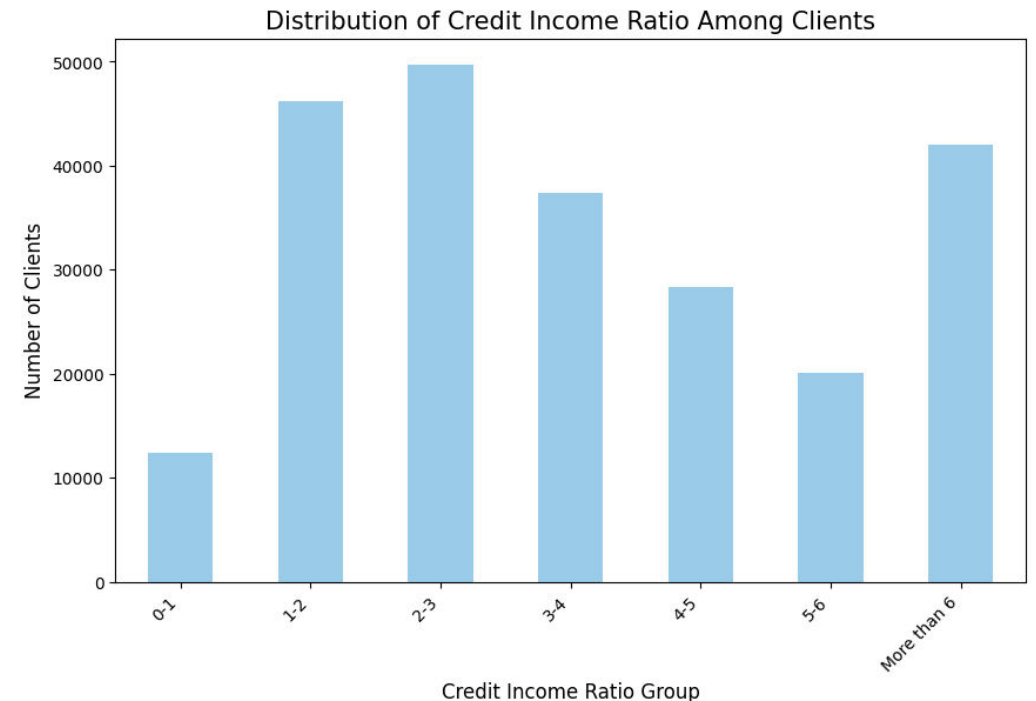
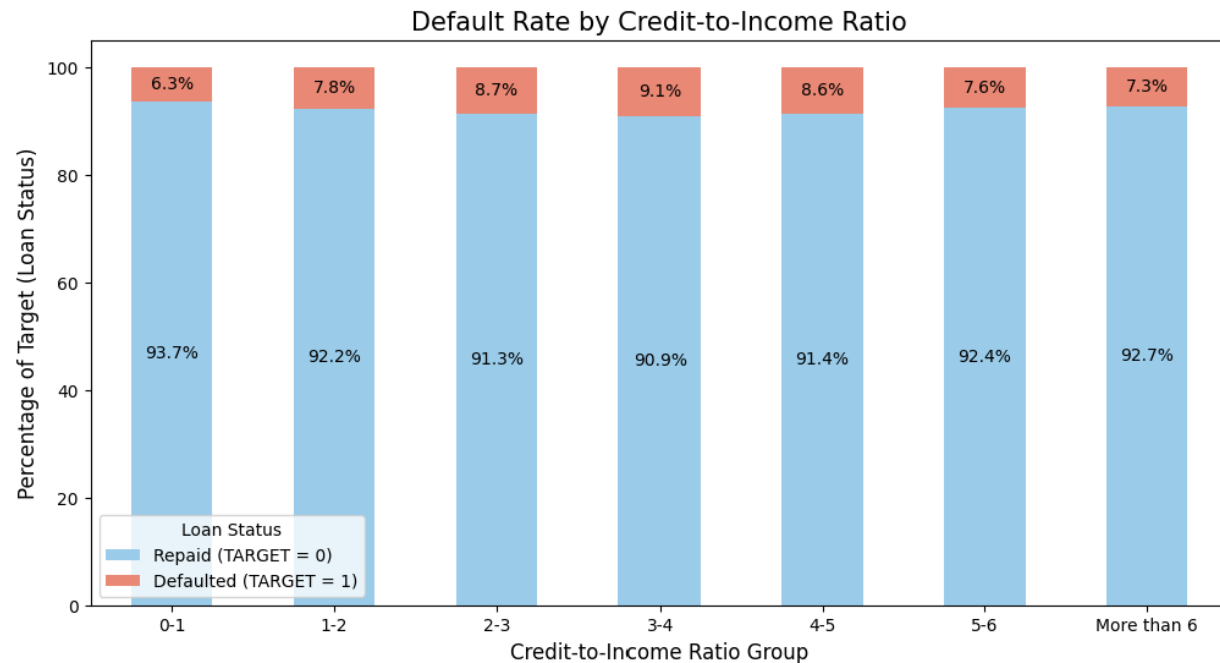


5. CATEGORICAL AND SEGMENTED UNIVARIATE ANALYSIS

CREDIT_INCOME_RATIO_GROUP Column

Insights

- 20-25 age group is the group having highest default rate it also the rationale for the number of clients distribution in this group is lower compared to other groups
- The correlation between default rate and the age group is parallel. The older they are the better default rate, especially with the group below 30 years old they have very high default rate.



5. CATEGORICAL AND SEGMENTED UNIVARIATE ANALYSIS

All of FLAG_DOCUMENT% Column

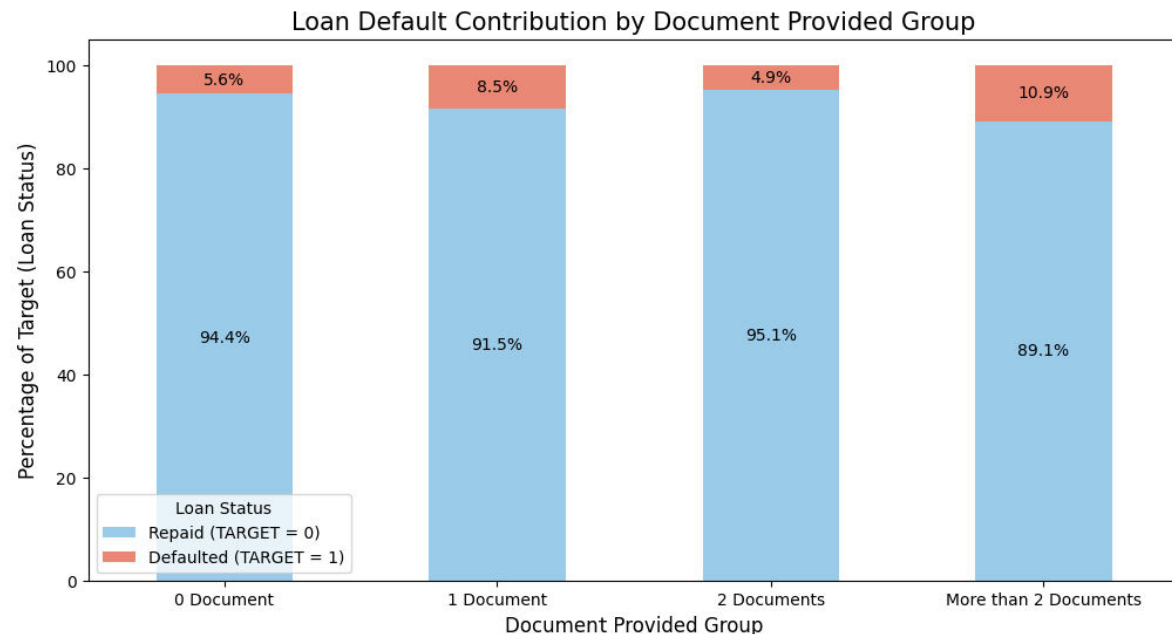
Problem: There are 20 columns of documents in which it is really to do the analysis document by document with the default rate

Solution:

- Step 1: Check the default rate of all FLAG_DOCUMENT columns in the table
- Step 2: Check how many clients provided in each document and add into the table below. In here, we only see document 3 and 6 having high distribution
- Step 3: Segment the document base on TOTAL_DOCUMENTS_PROVIDED has been declared from data quality check step and do the default rate analysis

Insights:

- The hypotheses would be the more document provided, the less default rate it is. However, there is the fluctuation of default rate among number of document provided
- Hence, those kind of columns will not be used for further analyses



```
DOCUMENTS_PROVIDED_GROUP
1 Document      207466
0 Document      22638
2 Documents     5952
More than 2 Documents 119
Name: count, dtype: int64
```

5. CATEGORICAL AND SEGMENTED UNIVARIATE ANALYSIS

FLAG_MOBIL, FLAG_EMP_PHONE, FLAG_WORK_PHONE, FLAG_PHONE, FLAG_EMAIL, TOTAL_CONTACT_METHODS Columns

Problem: There are 5 columns regarding the contact methods, and it cannot analyze by each method.

Solution:

- Combined each contact method and the TOTAL_CONTACT_METHODS columns create in the data quality check step to check default rate one time

Insights:

- Only FLAG_MOBIL having high default rate gap between 1 and 0 value. But after checking there is only 1 value in 0 so it also not make sense to make further analyses

	Column	Gap	Unique Values	Highest Default Rate
0	FLAG_MOBIL	8.105465	2	8.105465
5	TOTAL_CONTACT_METHODS	3.365139	5	9.096859
1	FLAG_EMP_PHONE	3.211285	2	8.683618
2	FLAG_WORK_PHONE	1.859855	2	9.594629
3	FLAG_PHONE	1.540319	2	8.538331
4	FLAG_EMAIL	0.206168	2	8.117160

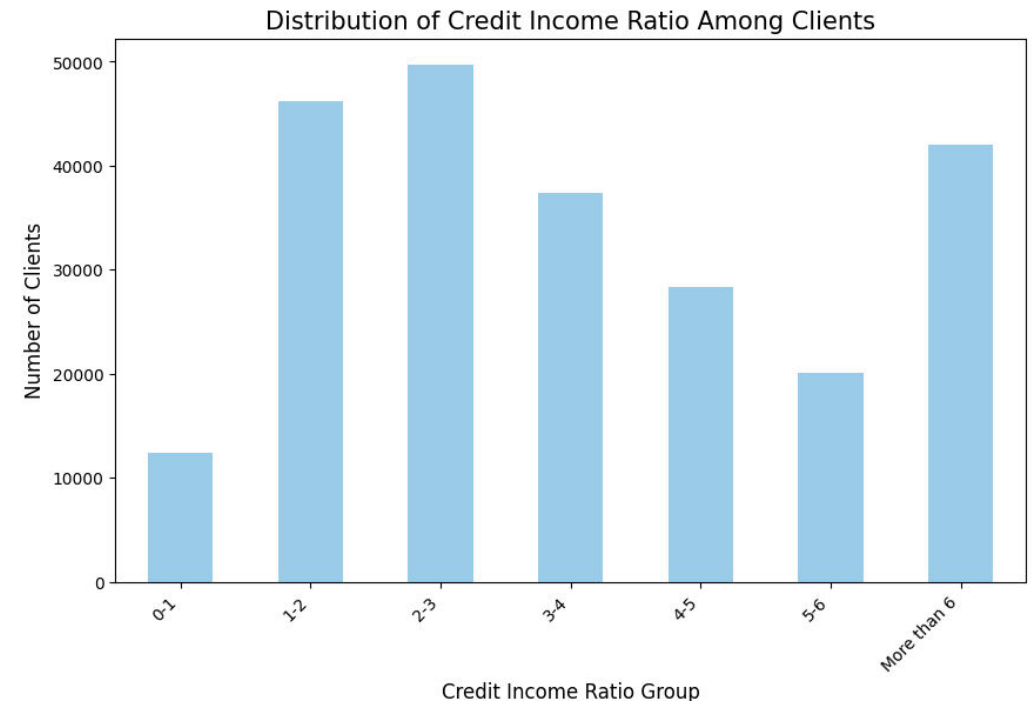
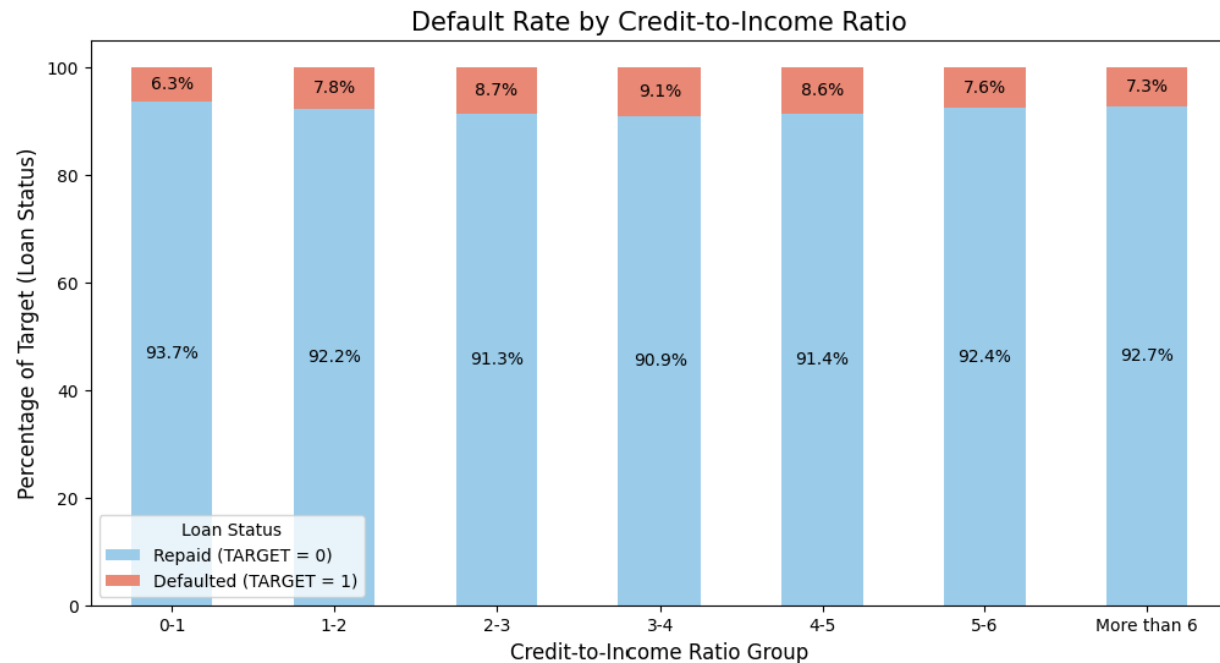
```
FLAG_MOBIL
1    236174
0         1
Name: count, dtype: int64
```

5. CATEGORICAL AND SEGMENTED UNIVARIATE ANALYSIS

CREDIT_INCOME_RATIO_GROUP Column

Insights:

- The group of ratio is having centralized in 3-4 with the highest default rate, and they do not have the correlation with the default rate while the lowest ratio and highest ratio having lowest default rate
- However, the gap is not significant, we can skip it and we can retest by the MONTHLYCREDIT_INCOME_RATIO_GROUP

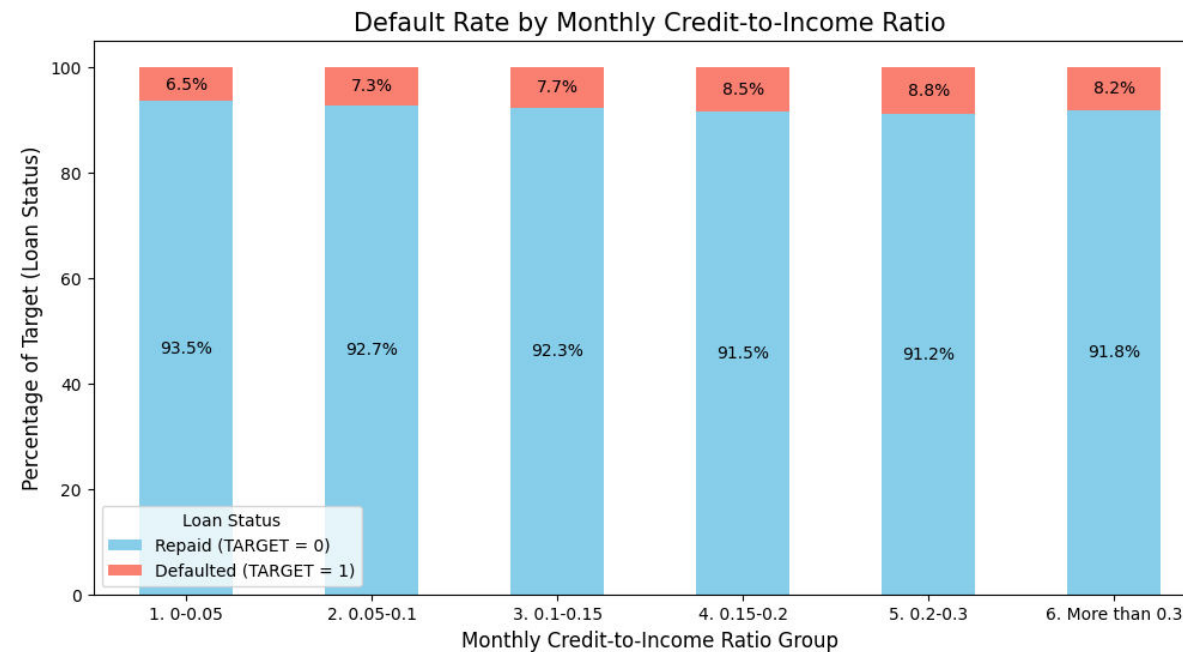


5. CATEGORICAL AND SEGMENTED UNIVARIATE ANALYSIS

MONTHLYCREDIT_INCOME_RATIO_GROUP Column

Insights:

- The group of ratio has shown that the more monthly payment they have to pay compared to their income, the more default rate the company is facing. It showed the correlation with default rate which can retest through bivariate analysis

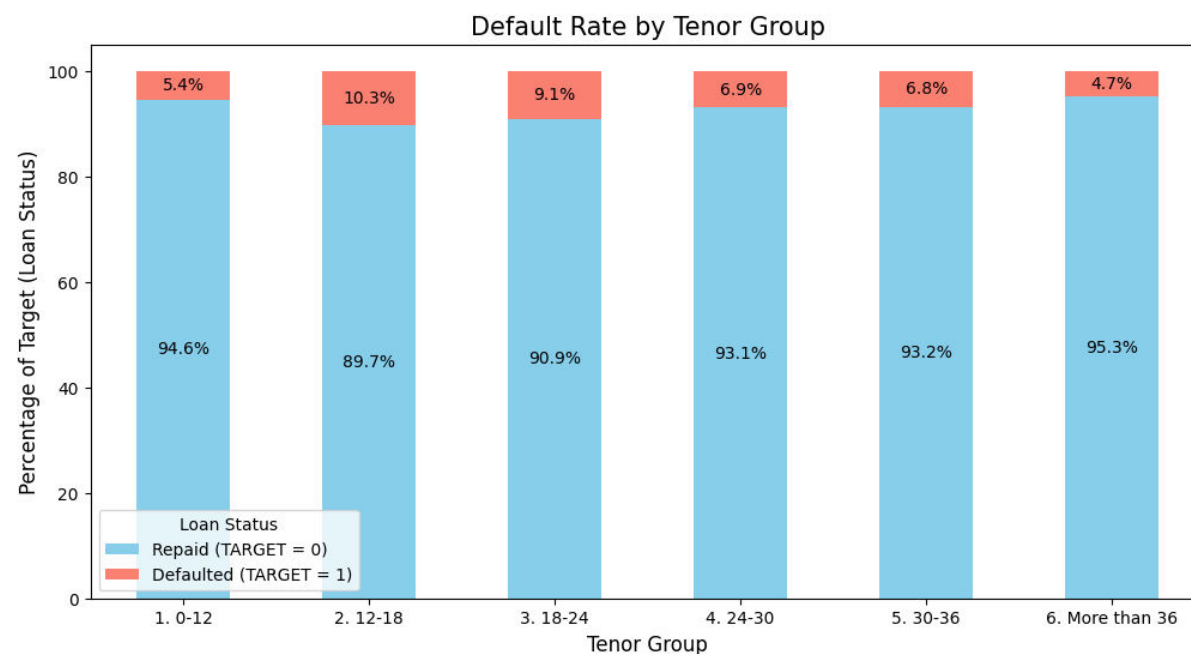


5. CATEGORICAL AND SEGMENTED UNIVARIATE ANALYSIS

TENOR_GROUP Column

Insights:

- The group of tenor have show surprised insight where tenor from 12 months upwards the higher tenor, the lower default rate which is correlate to the TARGET
- For the tenor 0-12, clients would look for the small amount of credit in which they are able to pay back faster also it is under their control of income; hence, the default rate in this tenor group is lower compared to the others



	TENOR_GROUP	AMT_ANNUITY
0	1. 0-12	27486.828232
1	2. 12-18	26172.424927
2	3. 18-24	24705.039684
3	4. 24-30	30325.554305
4	5. 30-36	28743.756213
5	6. More than 36	35827.795272

5. CATEGORICAL AND SEGMENTED UNIVARIATE ANALYSIS

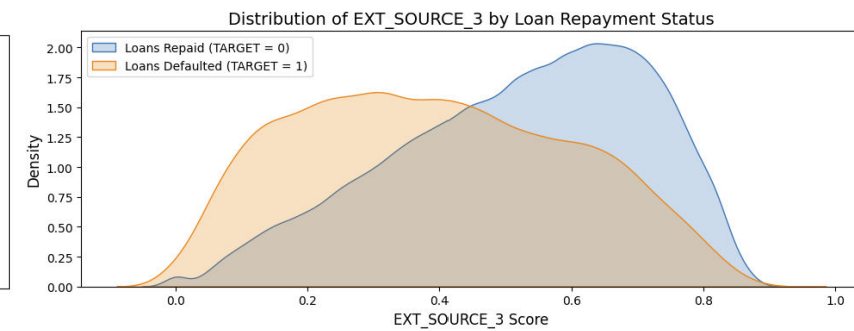
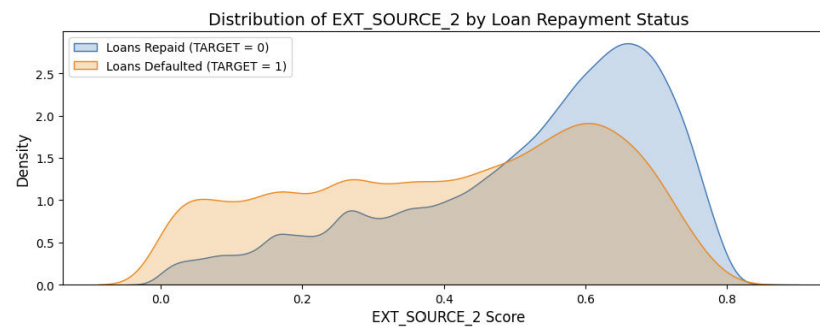
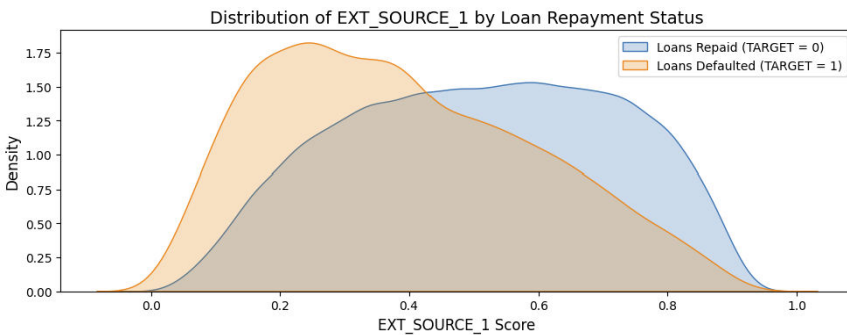
EXT_SOURCE% Columns

Problem: Not all of clients can get the data from external source

Solution: Accept the lost of data while testing the reaction between value in external source and the default rate

Insights: To make the insight clear we separate into two area, loan have default in orange area and loan prepaid in blue area

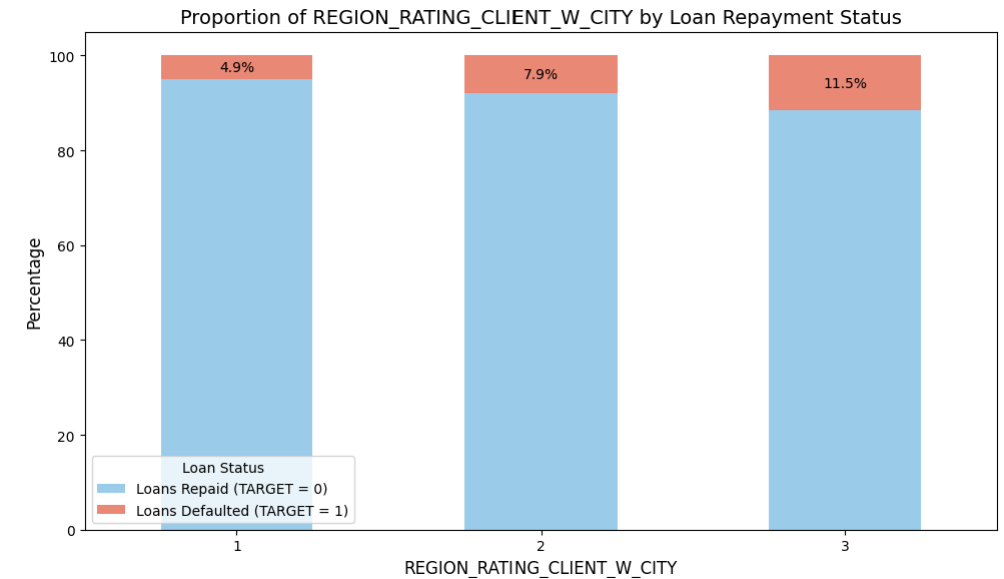
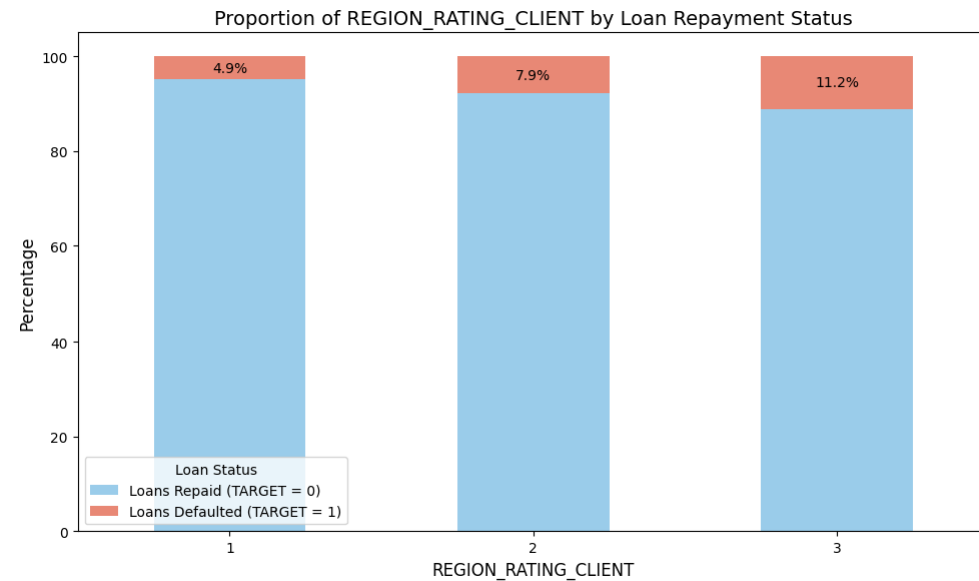
- The reaction of the default rate in External Source 1 and 3 is the same. In which, by providing lower score of external source the chance of loan default is higher and by providing lower score of external source the chance of loan prepaid is higher
- External Source 2 will not be used for further analysis as there is the positive correlation between prepaid and default loan among the rate in external source



5. CATEGORICAL AND SEGMENTED UNIVARIATE ANALYSIS

REGION_RATING_CLIENT, REGION_RATING_CLIENT_W_CITY Column

Insights: The ranking of region and city is heavily impacted on the default rate as the higher ranking (1->3) the higher default rate we observe. And the same trend for region itself and region with city



6. CORRELATION (NUMERIC) ANALYSIS



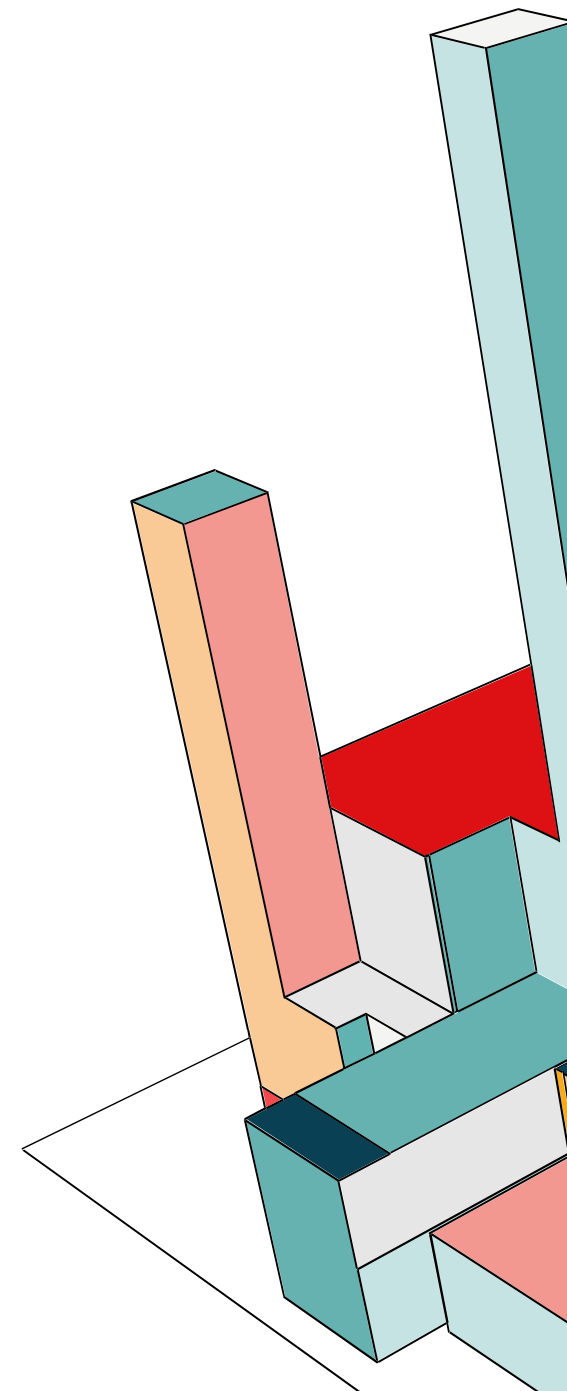
5. CORRELATION (NUMERIC) ANALYSIS

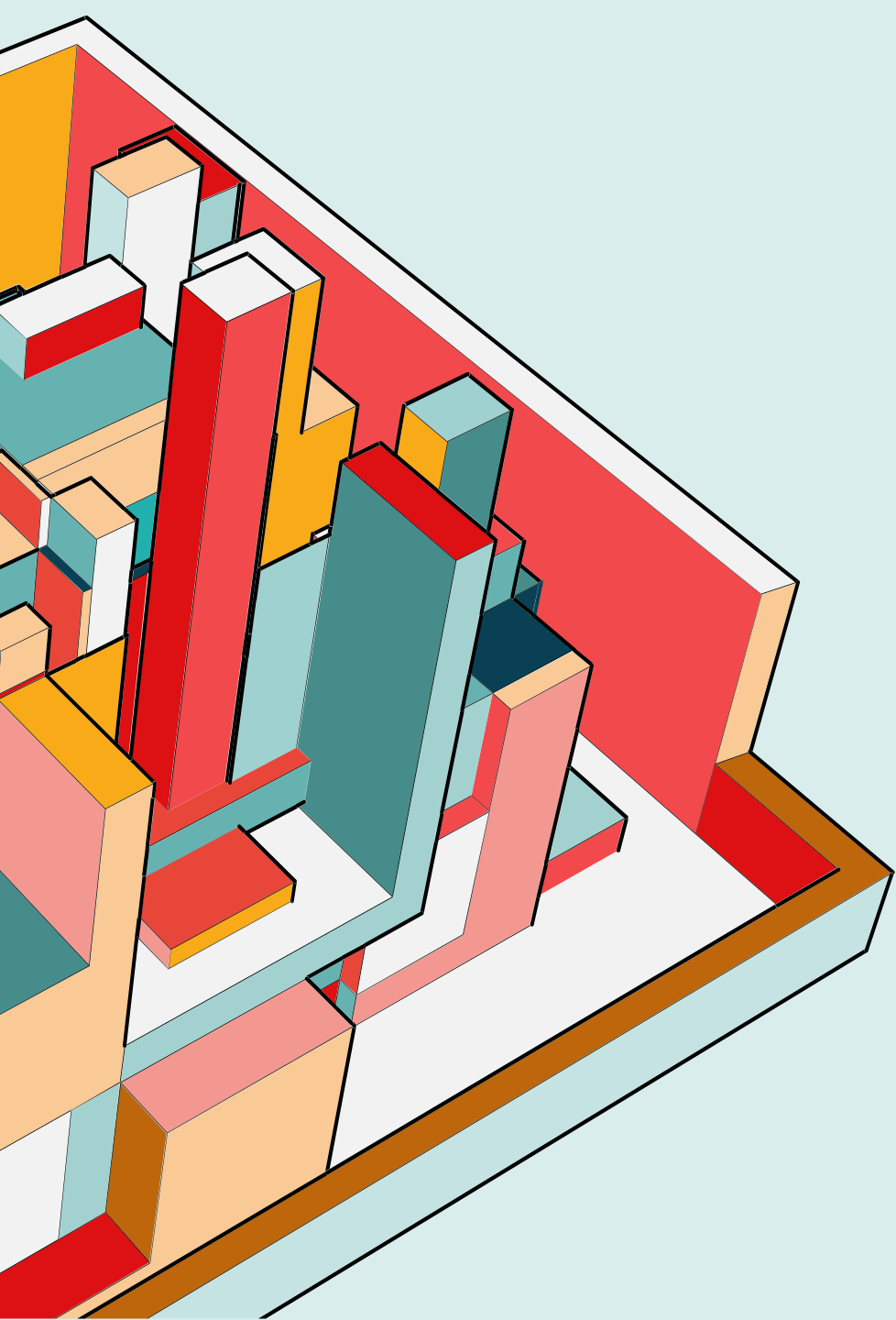
Top 10 correlations for clients with payment difficulties:

Factor 1	Factor 2	Correlation (%)
REGION_RATING_CLIENT	REGION_RATING_CLIENT_W_CITY	95.66%
REG_CITY_NOT_WORK_CITY	LIVE_CITY_NOT_WORK_CITY	77.86%
YEARS_EMPLOYED	AGE	65.38%
AGE	EXT_SOURCE_1	57.18%
REG_CITY_NOT_LIVE_CITY	REG_CITY_NOT_WORK_CITY	47.08%
EXT_SOURCE_1	YEARS_EMPLOYED	36.49%
YEARS_EMPLOYED	REG_CITY_NOT_WORK_CITY	27.05%
AGE	TENOR	21.39%
AGE	REG_CITY_NOT_WORK_CITY	21.32%
YEARS_EMPLOYED	LIVE_CITY_NOT_WORK_CITY	20.55%

Top 10 correlations for clients without payment difficulties:

Feature 1	Feature 2	Correlation (%)
REGION_RATING_CLIENT	REGION_RATING_CLIENT_W_CITY	94.99%
REG_CITY_NOT_WORK_CITY	LIVE_CITY_NOT_WORK_CITY	83.02%
YEARS_EMPLOYED	AGE	69.94%
AGE	EXT_SOURCE_1	60.09%
REG_CITY_NOT_LIVE_CITY	REG_CITY_NOT_WORK_CITY	43.69%
EXT_SOURCE_1	YEARS_EMPLOYED	38.37%
YEARS_EMPLOYED	REG_CITY_NOT_WORK_CITY	27.95%
REG_CITY_NOT_WORK_CITY	AGE	24.21%
LIVE_CITY_NOT_WORK_CITY	YEARS_EMPLOYED	22.65%
AGE	EXT_SOURCE_3	19.88%

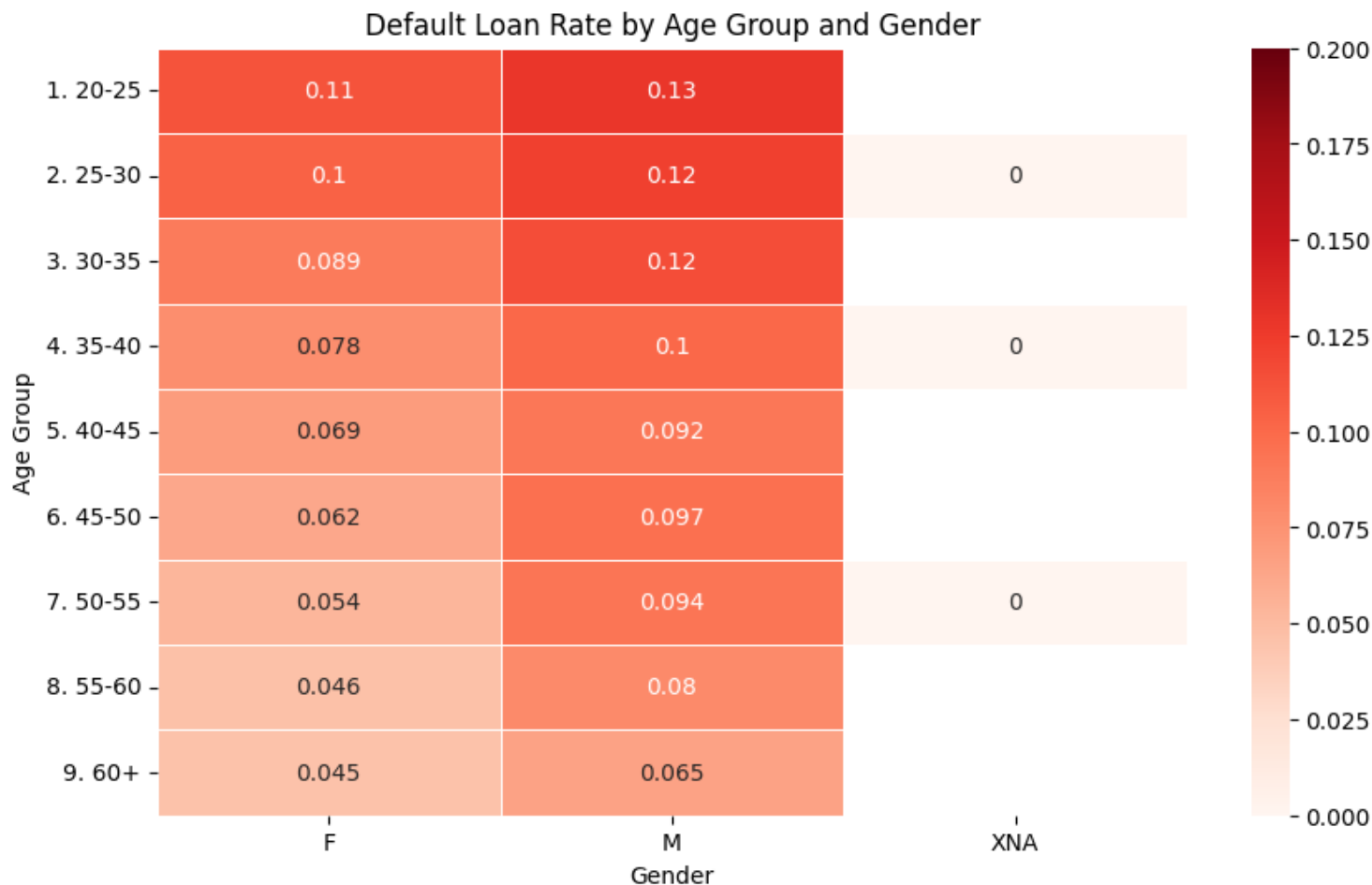




7. BIVARIATE ANALYSIS

BIVARIATE ANALYSIS

Heatmap Age Group and Gender



- The **highest default rates** were observed in younger clients:

- **Males (20-25 years):** 13%
- **Females (20-25 years):** 11%

Older clients (60+ years) tend to have the **lowest default rates**:

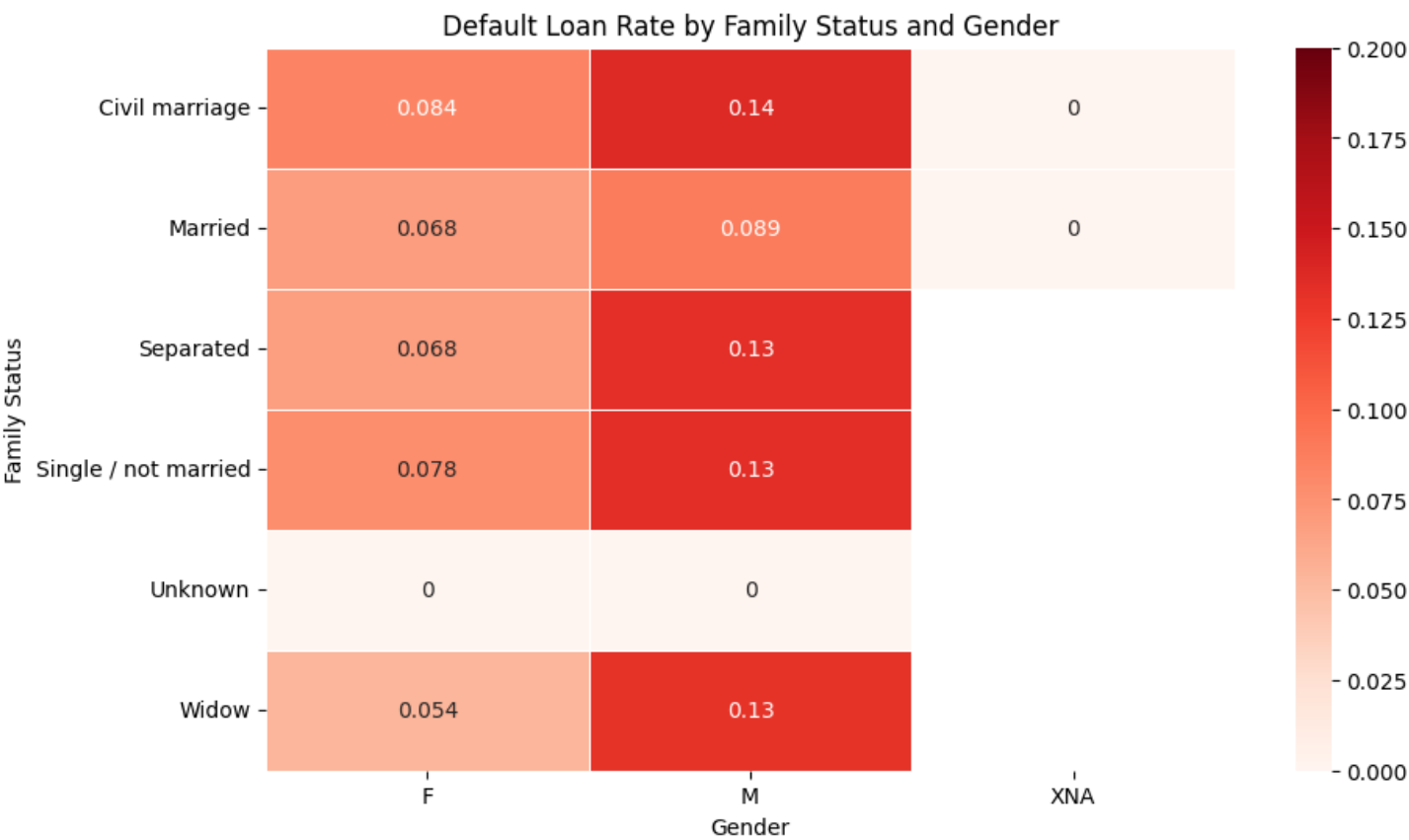
- **Females:** 4.5%
- **Males:** 6.5%

- The default rates for both males and females show a generally **declining trend with increasing age**, suggesting that older clients are less likely to default compared to younger ones

- **Males default rate always higher than Female** across the age group

BIVARIATE ANALYSIS

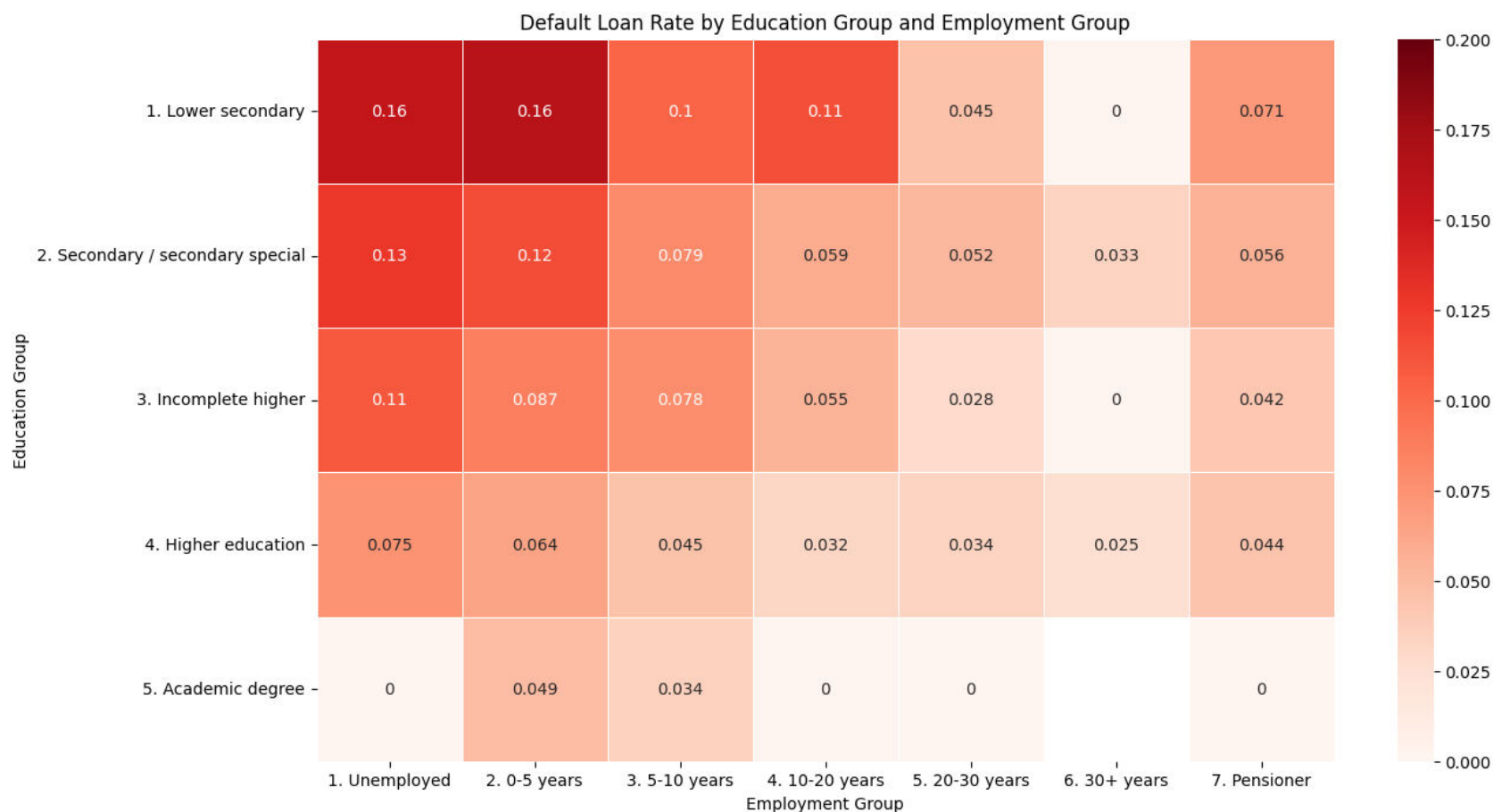
Heatmap Family Status and Gender



- Civil marriage has a higher default rate for males (14%) compared to females (8.4%)
- Civil Marriage, Separated, single, and widowed males tend to have higher default rates, and it is “stable” high at 13-14%
- The lowest default rate observed was for widowed females at 5.4%

BIVARIATE ANALYSIS

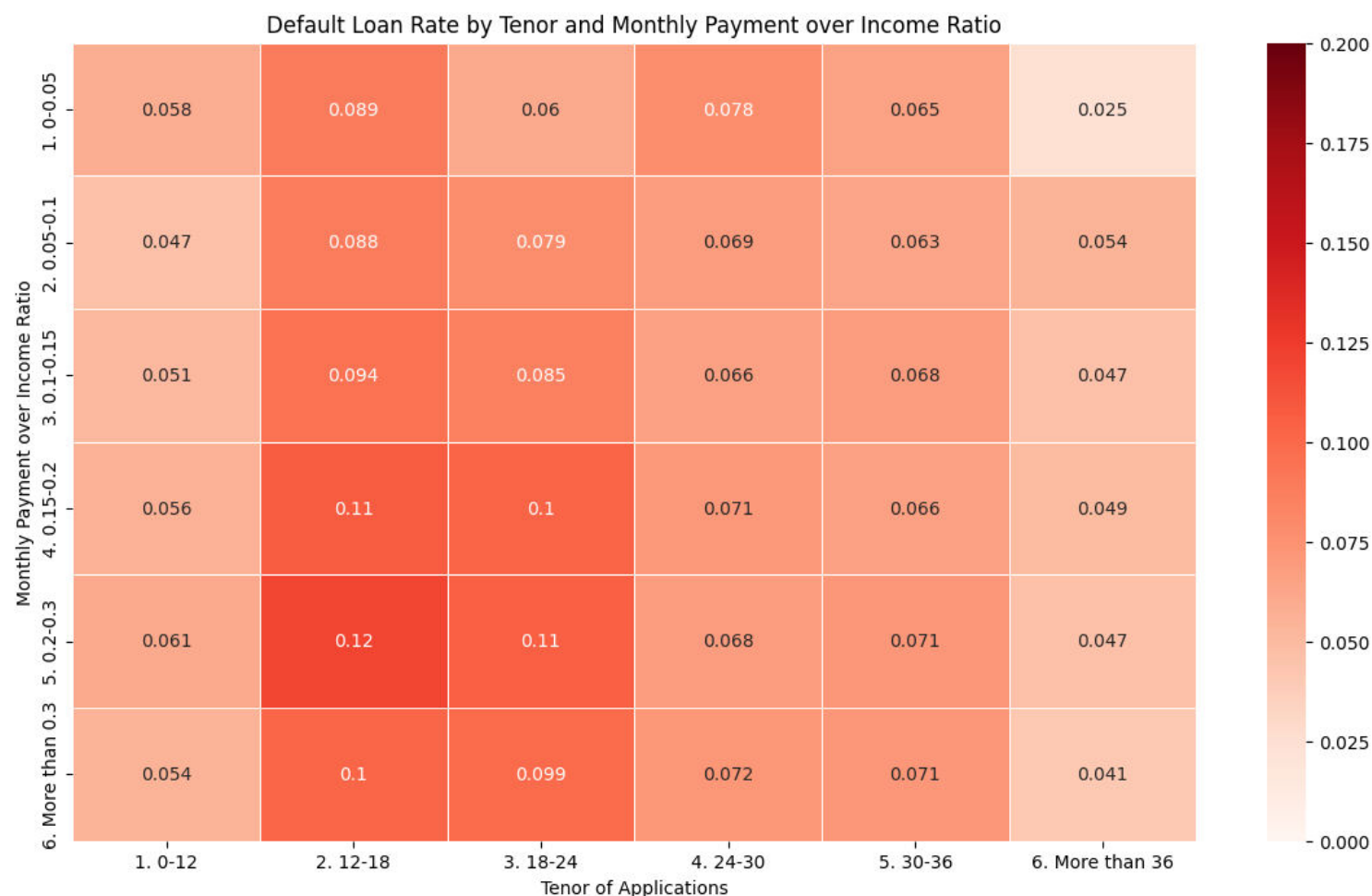
Heatmap Education and Employment Group



- **Unemployed group** except the academic decrease and higher education, it show the default rate quite high
- **Higher Education** and **Academic Degree** groups show significantly lower default rates across all employment groups. For those with an **academic degree**, the default rate is consistently **0%** for most groups, indicating strong financial stability
- **Lower Secondary Education** shows consistently high default rates across all employment groups, especially among **unemployed and 0-20 years employed group (>10%)**

BIVARIATE ANALYSIS

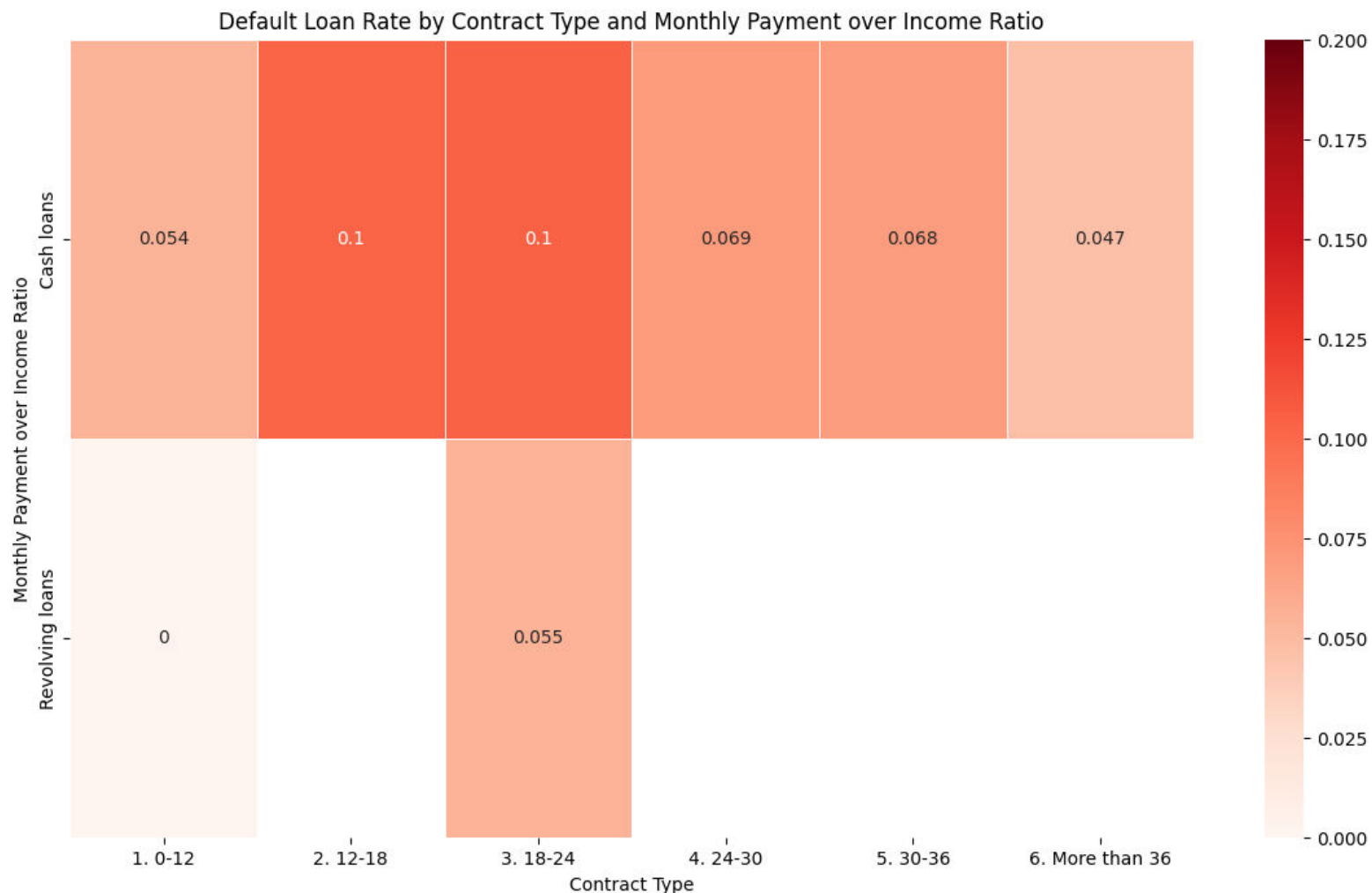
Heatmap Education and Employment Group



- **Tenor Group Effect::** Higher tenors (12-18 and 18-24 months) have higher default rates across all **monthly payment over income ratios** compared to shorter tenors
- **Longer Tenor (>36 months)** generally shows lower default rates compared to other tenors, suggesting clients opting for long-term tenors tend to be more financially stable
- **Shorter Tenor (<= 12 months)** have better default rate across the monthly payment over income ratio. Clients would not have much pressure on the shorter loan
- **The monthly payment over income ratio has a significant impact on the default rate.** Clients with a ratio of 0-0.05 and longer tenors have significantly lower default rates compared to other groups. And the correlation looks better for applications > 12 months

BIVARIATE ANALYSIS

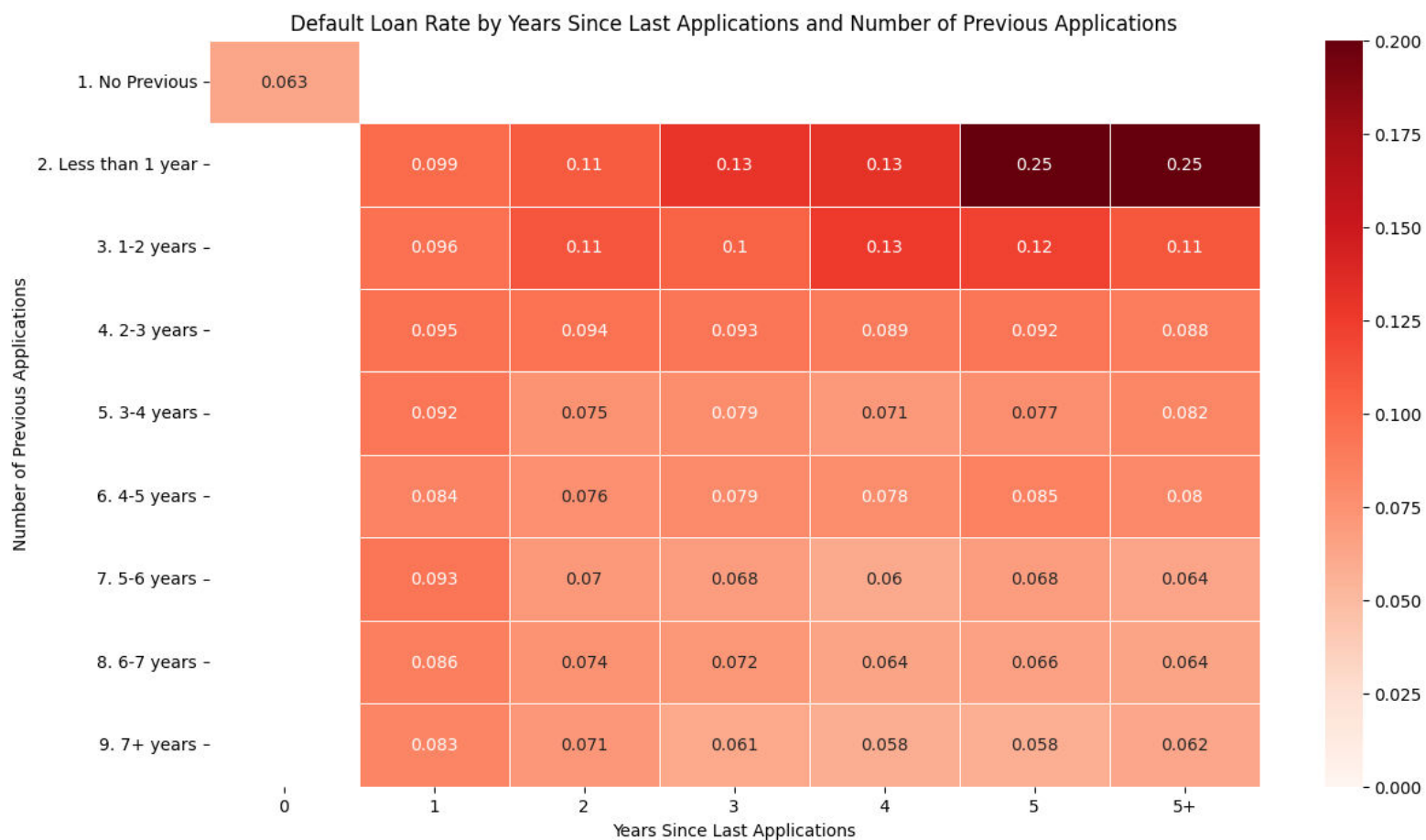
Heatmap Contract Type and Monthly Payment over Income Ratio



- **The Contract Type** influences default rates significantly, with **cash loans having higher default rates compared to revolving loans**, especially in shorter tenors (e.g., 12-18 and 18-24 months)
- **Default rates decrease with longer tenors for cash loans**, suggesting better loan performance as the repayment period extends

BIVARIATE ANALYSIS

Heatmap Years Since Last Applications and Number of Previous Applications



- High Default Risk for Short Time Since Last Application:

- The highest default rates (**up to 25%**) are observed for clients who applied less than a year ago
- Short intervals between previous applications seem to increase the risk

- Risk Reduction with Longer Intervals:

- The default risk decreases as the time since the last application increases
- Clients who have **7+ years since their last application** have significantly lower default rates (**5.8%-8.3%**)

- **Impact of Previous Applications:** Clients with no previous applications show the lowest default rate (**6.3%**), while those with multiple previous applications tend to have higher default rates

8. INSIGHTS SUMMARY

The analysis has provided several critical insights into the factors influencing loan default rates at Home Credit.
These insights are summarized in next slides



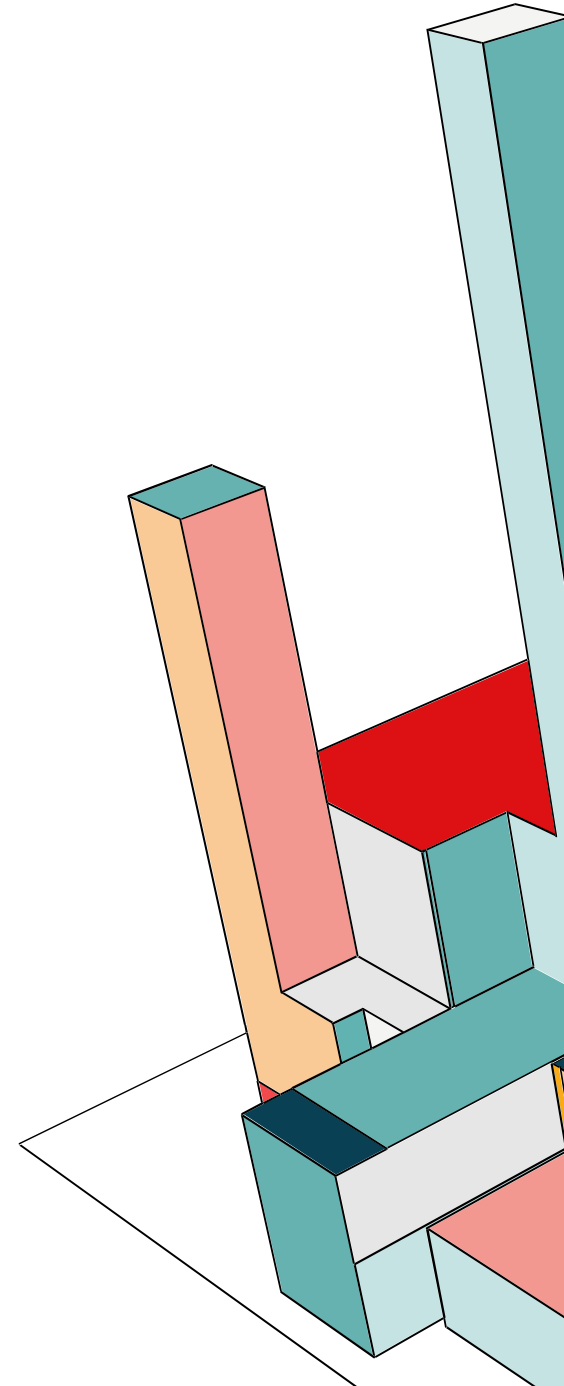
8. INSIGHTS SUMMARY

1. Demographic Factors:

- **Age and Gender:** Younger clients, particularly males aged 20-25, show the highest default rates. In contrast, older clients (60+ years) have the lowest default rates, especially among females. This suggests age is a significant factor in predicting loan repayment stability
- **Family Status:** Family status significantly impacts default rates. Clients in civil marriages or single have higher default rates, especially males, whereas widowed clients, especially females, show greater financial stability with lower default rates

2. Employment and Education:

- **Employment:** Employment status is a strong predictor of default risk. Unemployed clients have consistently higher default rates, while those with longer employment histories, especially pensioners, exhibit lower default risks
- **Education:** Education levels are directly linked to financial stability. Clients with higher education or academic degrees show significantly lower default rates, while those with only lower secondary education have consistently high default risks across employment groups



8. INSIGHTS SUMMARY

3. Loan and Contract Characteristics:

- **Tenor:** Loan tenor impacts default rates notably. Longer tenors (above 36 months) are linked to lower default rates, suggesting financially stable clients prefer longer repayment periods. Shorter tenors (12 months or less) also perform better, likely due to reduced financial strain. And it need to considered about the provide sustainable Tenor to have suitable EMI (Equated Monthly Installment) for clients
- **Contract Type:** Cash loans show higher default rates compared to revolving loans, particularly for shorter tenors. Extending the repayment period for cash loans improves loan performance

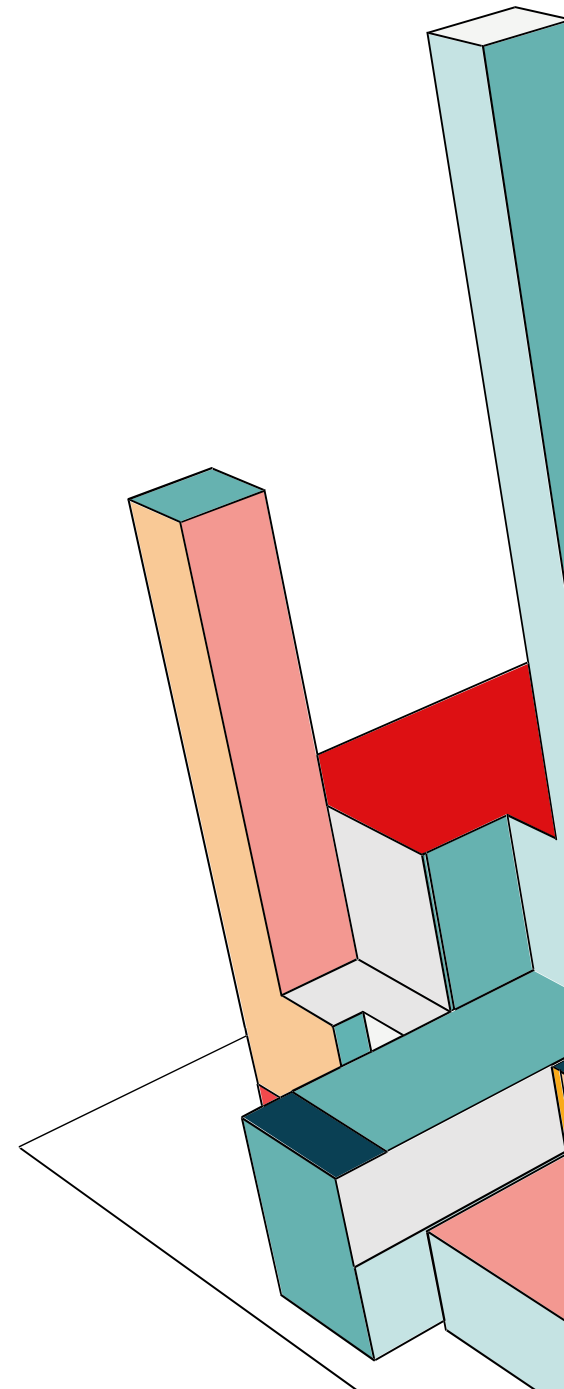
4. Income and Payment Ratios:

- **Monthly Payment-to-Income Ratio:** Higher monthly payment-to-income ratios are correlated with higher default rates, especially for clients with shorter tenors. Clients with lower ratios (0-0.05) and longer tenors exhibit significantly lower default rates
- **Credit-to-Income Ratio:** A higher credit-to-income ratio generally indicates a higher risk of default, although this correlation is not as strong as the monthly payment-to-income ratio

5. External Data Sources

- External Scores:** External scores, particularly EXT_SOURCE_1 and EXT_SOURCE_3, are effective in predicting default risk. Lower scores are associated with higher default rates. EXT_SOURCE_2, however, showed inconsistent correlations and may be less useful for further analysis.

6. Previous Application Behavior



8. INSIGHTS SUMMARY

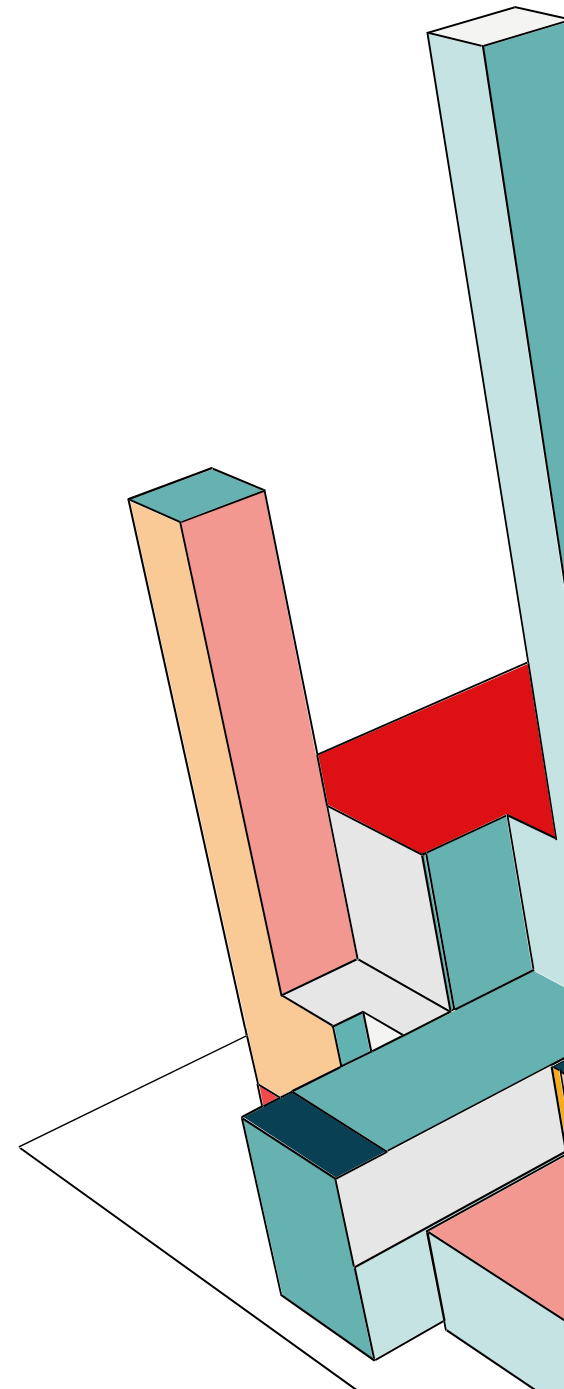
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6. Previous Application Behavior:

- **Recency of Applications:** Clients with recent previous applications (less than a year) have the highest default rates, up to 25%, the more clients apply in the shorter time the higher default risk potential it be. The risk of default decreases as the time since the last application increases, with clients who have not applied in over seven years showing the lowest default rates

- **Number of Previous Applications:** Clients with no previous applications have the lowest default rates, while multiple previous applications are associated with higher default risks. And this insight is need to be checked back as usually new client usually have higher risk portfolio as they do not have much data for observations

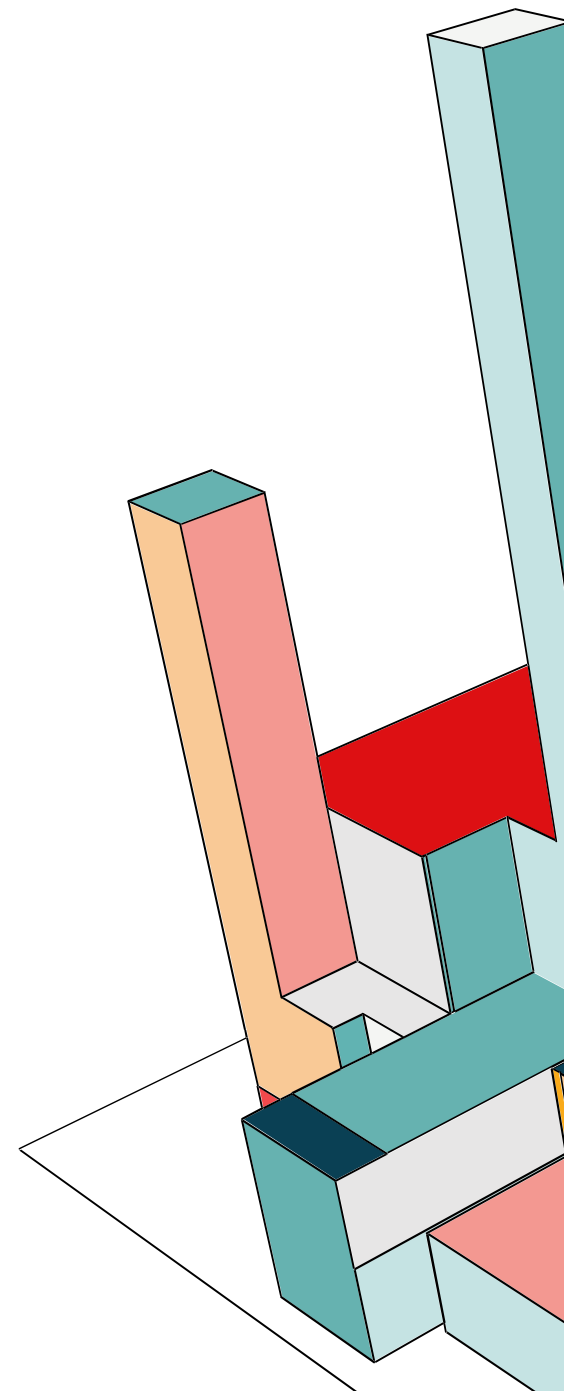


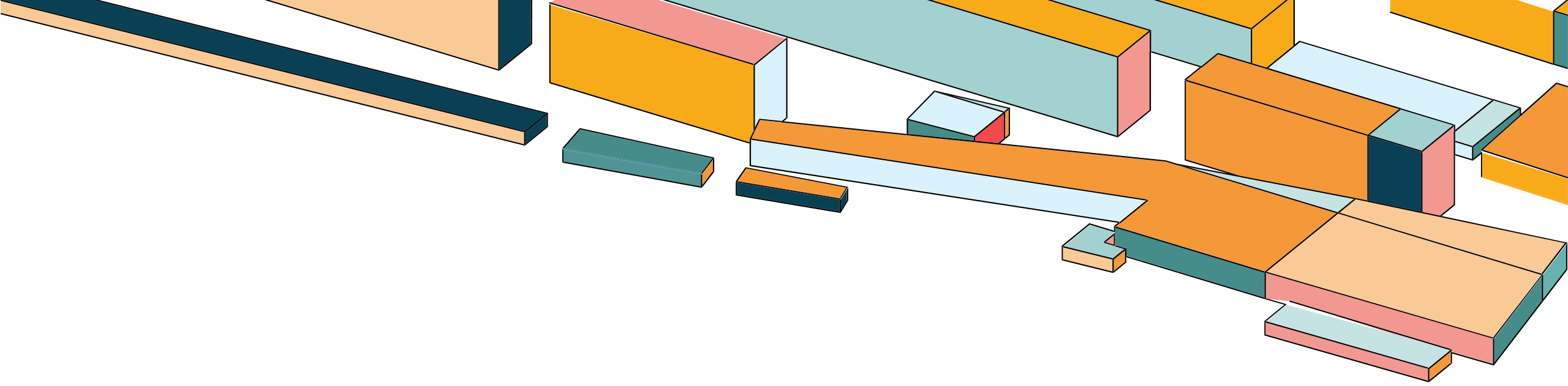
9. RECOMMENDATIONS



9. RECOMMENDATIONS

1. **Targeted Loan Offerings:** Prioritize offering longer tenors to clients who exhibit financial stability, such as those with **higher education levels** or **longer employment histories**. Avoid offering **short-tenor loans (less profit to company)** to clients with **high monthly payment-to-income ratios (high risk cost)**
2. **Enhanced Risk Assessment:** Integrate external scores (**EXT_SOURCE_1** and **EXT_SOURCE_3**) into the risk assessment models to enhance the accuracy of default predictions. Consider excluding EXT_SOURCE_2 due to its inconsistent correlations
3. **Customer Segmentation:** Segment clients based on demographic factors like **age, family status, and employment history** to offer customized loan products. Younger clients, particularly males, and those with lower education levels should be assessed with stricter criteria
4. **Monitor Previous Application Behavior:** Pay close attention to **clients with multiple recent applications**, as they have a **higher likelihood of default**. Implement stricter approval criteria or provide financial counseling for these clients
5. **Income Analysis:** Use income ratios to assess loan affordability more effectively. Clients with **high monthly payment-to-income ratios** should be offered **financial education** or **restructured loan terms (increase the tenor)** to reduce their risk of default.





THANK YOU

Duy Pham

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